



Digital Transformation in Strategic Project Management for Overcoming Execution Challenges in Multinational Engineering Firms

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ABSTRACT

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Engineering, Procurement and Construction (EPC) firms face growing execution challenges, particularly in cross-border environments marked by fragmented communication, cultural differences and supply-chain disruptions. This study investigates how digital transformation through technologies such as Artificial Intelligence (AI), Building Information Modelling (BIM), the Internet of Things (IoT) and automation can enhance project efficiency, transparency and resilience through improved supply-chain visibility and coordination.

The research focuses on six key organizational enablers: leadership commitment, cross-cultural collaboration, change management, digital skills and training, communication and transparency and governance support. A mixed-methods approach was applied, combining semi-structured interviews, a case study of CCC Engineering Ltd. (Bangladesh Operations) and a survey of professionals from Finland, Germany and Bangladesh.

The results show that technology alone does not ensure success. Leadership engagement, transparent communication and strong governance are the main drivers of digital transformation. Training and change management help balance adoption challenges, while supply-chain integration improves visibility and reliability.

The study presents a strategic framework for EPC firms, showing that digital transformation is an ongoing socio-technical process that aligns people, processes and technology. It highlights that leadership, transparent communication and continuous learning are essential to achieving sustainable efficiency and trust across global projects.

The findings provide both academic and practical value by linking technology adoption with human and organizational readiness. The research contributes to developing resilient, transparent and future-ready project management practices for the global EPC industry.

Key words: digital transformation, strategic project management, supply chain, EPC, leadership, systems thinking, cross cultural collaboration, change management.

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GLOSSARY or ABBREVIATIONS AND TERMS

Abbreviation / Term	Definition / Description
AI	Artificial Intelligence- Simulation of human intelligence in machines, used in data analysis, automation and predictive modelling.
BIM	Building Information Modelling- A digital representation of the physical and functional characteristics of a facility.
CCC Engineering Ltd.	Case company used in this study; A multinational EPC contractor specializing in power transmission projects.
EPC	Engineering, Procurement and Construction- A common project delivery model in large infrastructure and industrial projects.
IoT	Internet of Things- A network of interconnected devices enabling data exchange for monitoring and control.
KPI	Key Performance Indicator- Measurable metric used to evaluate performance and progress toward project goals.
PMBOK	Project Management Body of Knowledge Standardized framework of project management best practices developed by PMI.
SC	Supply Chain Network of organizations, people and processes involved in delivering a project or product from source to customer.
TAM	Technology Acceptance Model- Theoretical model explaining user acceptance of technology based on perceived usefulness and ease of use.
TAMK	Tampere University of Applied Sciences- Degree awarding institution of this master's thesis.
Systems Thinking	Holistic analytical approach emphasizing interconnections, feedback loops and dynamic relationships within complex systems.
Governance	The framework of rules, policies and processes that ensure accountability, transparency and compliance in digital transformation.
Change Management	Structured approach to transitioning individuals, teams and organizations to desired future states during transformation.

1 INTRODUCTION

The Engineering, Procurement and Construction (EPC) sector is a cornerstone of global infrastructure development, enabling the delivery of energy plants, transportation networks and industrial facilities across national borders. These projects are typically characterized by high complexity, multi-stakeholder involvement and substantial financial commitments. Despite decades of accumulated project management knowledge, the industry continues to face persistent challenges in delivering projects on time and within budget. Global surveys indicate that over 60% of large EPC projects experience cost overruns and delays (Project Management Institute, 2023), with cross border projects being particularly vulnerable due to cultural, logistical and coordination difficulties.

The emergence of the fourth industrial revolution offers significant opportunities to address these execution challenges. Digital technologies such as artificial intelligence (AI), blockchain, cloud-based collaboration platforms and real-time analytics promise to transform how projects are planned and executed. AI can enhance predictive decision making and automate repetitive tasks; blockchain can secure procurement processes and improve supply chain transparency and cloud systems can foster collaboration across geographically dispersed teams. However, adoption of these tools within EPC firms has been inconsistent, often limited to pilot projects or isolated applications rather than systemic integration (Chen, 2025; Musaigwa & Swanepoel, 2025).

According to the Project Management Institute (2023), more than 60% of large-scale EPC projects globally fail to meet schedule or budget targets, with cost overruns averaging 30%. Recent McKinsey reports (2024) highlight that while over 70% of EPC firms recognize digital transformation as a strategic priority, fewer than 30% have successfully scaled digital initiatives beyond pilot projects.

Global disruptions such as COVID-19, geopolitical instability and rising sustainability requirements have intensified these challenges. At the same time, stricter environmental, social and governance (ESG) reporting standards have increased the pressure on EPC firms to adopt technologies that enhance transparency, resilience and efficiency across supply chains.

Digital transformation is increasingly viewed not only as a means of operational efficiency but also as a catalyst for sustainable strategic project management. For example, IoT systems allow real-time tracking of energy consumption and emissions across project sites, while AI driven analytics support the optimization of resource allocation to minimize carbon footprints. Similarly, blockchain based traceability systems enhance accountability in sourcing sustainable materials. Industry research shows that EPC firms embedding sustainability within their digital strategies are better positioned to secure contracts, meet regulatory standards and maintain stakeholder trust (Sundarakani, 2024; Harvard Business Review, 2023).

Thus, digital transformation represents both an operational and a strategic pathway for EPC firms seeking to improve performance and meet global sustainability targets.

This study specifically investigates six organizational enablers namely leadership commitment, cross cultural collaboration, change management, digital skills and training, communication and transparency and governance support that collectively shape digital transformation outcomes in EPC projects.

2 PROBLEM STATEMENT

While digital transformation has attracted significant attention in both industry and academia, most Engineering, Procurement and Construction (EPC) firms still struggle to translate its potential into measurable execution improvements. Previous research has often examined digital technologies in isolation (for example, AI adoption or blockchain in supply chains) without sufficiently addressing how these tools interact with organizational factors such as leadership commitment and cross cultural collaboration. At the same time, project management research has long recognized that human and cultural dimensions are as critical as technical solutions. This fragmentation leaves a clear gap: limited understanding exists of how digital transformation can be systematically aligned with both the technological and organizational realities of multinational EPC projects.

Existing studies on digital transformation in EPC remain fragmented, with most focusing narrowly on individual technologies such as AI or blockchain (Jing & Fan, 2024; Chen, 2025). However, limited scholarship integrates these technological advances with organizational enablers like leadership, governance and cross-cultural collaboration. As a result, theory often lags behind practice, leaving EPC firms without comprehensive guidance for implementing digital transformation at scale.

The consequences of this fragmented approach are evident in practice. Many EPC firms invest in isolated digital tools without cohesive integration strategies, leading to duplicated systems, inconsistent data flows and low returns on investment. Recent PMI (2023) findings reveal that nearly 70% of digital transformation projects in project-based industries fail to progress beyond pilot stages or produce measurable value. In EPC environments, this results in persistent delays, supply chain inefficiencies and underutilization of advanced technologies that could otherwise improve project reliability and transparency.

Therefore, this research addresses the critical need to bridge the gap between technology and organization in EPC project execution. It examines how six key enablers collectively influence the successful alignment of digital transformation initiatives with project performance outcomes.

3 LITERATURE REVIEW

3.1 Research Objectives

The primary aim of this research is to develop a strategic digital transformation framework that supports successful project execution in multinational EPC firms.

The specific objectives are:

- Identify and categorize execution challenges in multinational EPC firms.
- Evaluate the effectiveness of digital transformation tools (AI, blockchain, cloud solutions) in overcoming these challenges.
- Examine how organizational enablers such as leadership commitment, cross cultural collaboration, change management, digital skills and training, communication transparency and governance support influence the success of digital transformation initiatives.
- Develop a strategic framework for digital transformation tailored to multinational project environments.
- Benchmark best practices from leading EPC firms.
- Provide actionable recommendations for managers and policymakers.

3.2 Research Questions

To operationalize these objectives, the study is guided by two central research questions:

1. *How can digital technologies such as AI, BIM, IoT and blockchain enhance supply chain efficiency, transparency and overall project performance in multinational EPC projects?*

This question is addressed primarily through a quantitative survey of EPC professionals and a case study of CCC Engineering Ltd. The analysis combines descriptive and correlational methods to measure the perceived impact of digital tools on project efficiency, transparency and resilience.

2. *In what ways do organizational enablers including leadership, collaboration, change management, digital skills, communication and governance influence digital transformation adoption and success in multinational EPC firms?*

This question is examined through qualitative analysis of ten semi structured interviews and quantitative validation from survey results. The analysis uses thematic coding and systems thinking synthesis to identify how leadership, culture and governance practices shape successful digital transformation outcomes.

Research Question 1 focuses on the technological dimension of digital transformation and is primarily addressed through quantitative analysis of survey data and case-study evidence. Research Question 2 examines the organizational enablers of transformation, analyzed through thematic coding of interviews and quantitative correlation testing, with both questions integrated using a systems thinking framework.

3.3 Justification and Significance of the Study

This research is significant for both practice and theory. Practically, EPC firms require evidence based guidance on how to leverage digital tools to address execution bottlenecks. By focusing on supply chain digitalization, this study targets one of the most delay prone and risk-intensive areas of EPC projects. At the same time, it emphasizes that technology adoption is inseparable from leadership support and cultural alignment, ensuring that organizational realities are not overlooked. Academically, this study contributes by integrating the technological perspective (digital tools and supply chain systems) with the organizational perspective (leadership, collaboration and communication) into a single framework, addressing a gap that current literature treats separately (Reis, 2023; Jing & Fan, 2024; Sundarakani, 2024).

From an academic standpoint, this thesis advances knowledge by integrating multiple perspectives into a single coherent framework. While the Technology Acceptance Model (TAM) explains user level adoption and Systems Thinking highlights interdependencies across organizations, PMBOK provides standardized project management practices. By combining these three approaches, this study contributes a unique theoretical model tailored to the complexities of EPC projects. It extends digital transformation research beyond isolated technology studies by positioning digital adoption as a sociotechnical process, where technological innovation and organizational enablers are inseparable.

From a managerial perspective, the study is highly relevant to EPC practitioners who face increasing pressure to deliver projects efficiently while managing global supply chains and multicultural teams. The framework developed here will provide actionable insights for project managers and executives seeking to scale digital transformation initiatives across borders. In particular, it will offer guidance on how to integrate AI, blockchain and automation into supply chain processes

while simultaneously building leadership commitment and fostering cross cultural collaboration.

The research also carries policy significance. As governments and industry bodies introduce regulations and standards for digital adoption in infrastructure and construction, evidence based frameworks are required to guide these transitions. The findings of this study may inform policymakers on the conditions under which digital transformation leads to improved project outcomes, thereby shaping standards for transparency, accountability and sustainability. By linking technological adoption to both organizational readiness and regulatory environments, the study supports not only academic and managerial audiences but also the broader ecosystem in which EPC projects operate.

3.4 Scope and Limitations

The research focuses on multinational EPC firms and their project execution practices. Supply chain management is treated as the primary area for applying digital transformation, while leadership and cross cultural aspects are analyzed as contextual enablers. Limitations include the scope of case studies (e.g., CCC Engineering Ltd.), the sample size of industry interviews and surveys and the challenge of generalizing findings beyond the selected firms.

The scope of this study is limited to multinational EPC firms operating across Europe, Asia and the Middle East, where cross cultural collaboration and extended supply chains are most prominent. The technological scope focuses primarily on AI, blockchain, automation, IoT and Building Information Modelling (BIM), which are the most widely discussed tools in the literature and industry reports between 2023 and 2025. Leadership and cross cultural collaboration are included as organizational enablers rather than as independent constructs. Limitations include the reliance on selected case studies and a relatively small sample size of industry professionals, which may restrict the generalizability of findings.

In terms of temporal scope, this research focuses on developments in the period 2023–2025, a phase marked by accelerated digital adoption in the post COVID recovery era, heightened sustainability requirements and continued global supply chain disruptions. By concentrating on this period, the study ensures that the findings reflect the most recent technological and organizational trends affecting EPC firms.

Geographically, the scope is limited to multinational EPC firms operating across Europe, Asia and the Middle East regions where cross border collaboration and cultural diversity are most evident in project execution. This ensures that the research captures contexts where digital transformation is not only a technological issue but also a matter of managing global coordination and diverse cultural perspectives.

In terms of technological scope, the study emphasizes AI, blockchain, automation, IoT and Building Information Modelling (BIM). These technologies have been consistently highlighted in recent literature and industry reports as the most influential in shaping EPC project execution. Other technologies such as augmented reality (AR) or virtual reality (VR) are acknowledged but excluded from detailed discussion, as their application in EPC project management remains limited.

Finally, the study acknowledges certain limitations. The reliance on a relatively small sample of interviews and surveys, along with a case study focus on CCC Engineering Ltd., may limit the generalizability of findings. Moreover, small subcontractors and non project based firms are outside the scope, as the study targets large multinational EPC firms where execution challenges and digital transformation initiatives are most pronounced.

3.5 Structure of the Thesis

This thesis is organized into eight chapters. Chapter 1 introduces the research background and sets the context. Chapter 2 defines the problem statement, while Chapter 3 reviews the literature to establish justification, objectives and scope. Chapter 4 develops the theoretical framework, drawing on digital transformation literature, supply chain studies and project management models such as TAM, Systems Thinking and PMBOK. Chapter 5 describes the research methodology, including the mixed methods design, data collection and analysis procedures. Chapter 6 presents the empirical findings, followed by Chapter 7, which discusses their implications in relation to existing literature and practice. Finally, Chapter 8 concludes the thesis with a summary of key findings and reflections and offers recommendations for industry practice and future research.

3.6 Chapter Summary

This chapter introduced the context, motivation and purpose of the study. It outlined how multinational EPC firms face persistent execution challenges, including fragmented coordination, supply chain disruptions and cultural complexities, despite the growing availability of digital technologies. The problem statement identified a clear gap in both practice and research: while technologies such as AI, blockchain, BIM and IoT offer significant potential, their adoption often fails without organizational enablers such as leadership commitment, cross cultural collaboration, structured change management and digital skills development.

The research objectives and questions were defined to address this gap by developing a strategic digital transformation framework that integrates technological and organizational dimensions. The justification highlighted the academic significance of combining TAM, Systems Thinking and PMBOK into one coherent approach, while also emphasizing the practical value for EPC managers, policy makers and industry stakeholders. The scope clarified the focus on multinational EPC firms, key technologies and the 2023–2025 timeframe, while acknowledging limitations in generalizability due to sample size and case study design.

Overall, this chapter established the foundation of the thesis by positioning digital transformation as a **socio-technical process** that requires both technological innovation and organizational readiness to achieve meaningful improvements in project execution. The next chapter builds on this foundation by reviewing the theoretical frameworks and existing literature, providing the conceptual grounding for the empirical research that follows.

4 THEORETICAL FRAMEWORK

4.1 Introduction to the Theoretical Framework

This chapter establishes the theoretical foundation of the study. It integrates perspectives from digital transformation research, supply chain management literature and organizational theories, supported by established project management models. This combined lens is essential because the execution challenges in multinational EPC projects stem from the interaction between technical systems, organizational processes and human behavior.

Digital transformation has been widely recognized as a strategic driver of competitiveness in industry, yet its role in project based contexts remain underexplored (Reis, 2023). While technologies such as artificial intelligence (AI), blockchain and cloud platforms can significantly improve efficiency, their adoption in EPC firms has been fragmented and often fails to deliver expected outcomes. Scholars argue that this gap arises from the absence of alignment between digital initiatives and the practical realities of project execution (Chen, 2025). The framework developed in this thesis therefore builds on both technological enablers (AI, blockchain, automation) and organizational enablers (leadership and cross cultural collaboration) to address this dual challenge.

4.2 Digital Transformation in Project-Based Contexts

Digital transformation is defined as the integration of digital technologies into business processes to fundamentally reshape operations, decision-making and value creation (Westerman, Bonnet & McAfee, 2014). Unlike conventional IT adoption, digital transformation involves redesigning workflows, strategies and organizational culture.

In project-based industries such as EPC, the stakes of transformation are particularly high. Projects are temporary organizations that require intensive coordination among stakeholders, suppliers and contractors across international boundaries. Recent reviews emphasize that digital transformation in EPC must be embedded into strategic project management frameworks rather than treated as standalone IT upgrades (Reis, 2023). Musaigwa and Swanepoel (2025), in their systematic review of industrial engineering, highlight that firms gain measurable

improvements only when digital initiatives are integrated with operational practices such as supply chain workflows.

At the same time, EPC projects face cultural and organizational challenges that complicate adoption. Chen (2025) notes that while AI and blockchain offer significant potential for predictive analytics and transparency, successful implementation depends on organizational readiness and leadership governance. This underscores the sociotechnical nature of digital transformation: technology adoption cannot succeed without addressing human and managerial dimensions.

4.3 Key Technologies in Digital Transformation

4.3.1 Artificial Intelligence (AI)

AI is increasingly applied in project management for predictive analytics, anomaly detection and real-time decision support. In EPC supply chains, AI enables demand forecasting, supplier risk assessment and optimization of logistics networks. Chen (2025) demonstrates that AI driven decision support systems reduce inefficiencies by providing early warnings of schedule deviations and procurement risks. Similarly, Jing and Fan (2024) show that AI enabled supply chain integration improves coordination between manufacturers and contractors, leading to measurable gains in efficiency and transparency.

4.3.2 Blockchain

Blockchain has emerged as a promising tool for addressing procurement risks in EPC supply chains. By providing a decentralized, immutable ledger of transactions, blockchain enhances transparency and reduces fraud. Jing and Fan (2024) provide empirical evidence from manufacturing supply chains showing that blockchain reduces procurement delays and improves supplier accountability. IEEE studies (2022–2024) further highlight its potential for contract automation and dispute reduction in project management. However, adoption is still limited due to technical complexity and lack of cross industry standards.

4.3.3 Automation and Robotics

Automation, including robotic process automation (RPA), streamlines repetitive tasks such as contract verification, reporting and compliance monitoring. Mu-saigwa and Swanepoel (2025) identify RPA as a critical enabler of supply chain

optimization, reducing administrative bottlenecks that often slow down project delivery. Harvard Business Review (2023) emphasizes that automation must be implemented alongside upskilling programs to prevent resistance and ensure sustainable adoption in complex organizations.

4.3.4 Internet of Things (IoT) and Digital Twins

IoT technologies facilitate real-time tracking of equipment, materials and logistics flows, which is vital for EPC projects with extended supply chains. Sundarakani (2024) argues that IoT and digital twins significantly enhance supply chain resilience and sustainability, enabling predictive maintenance and reducing downtime risks. Reis (2023) further notes that while IoT adoption has grown rapidly, integration challenges persist due to fragmented data architectures and cybersecurity risks.

4.3.5 Building Information Modelling (BIM)

Building Information Modelling (BIM) has become a central digital tool in engineering and construction projects, enabling the creation of shared 3D models that integrate design, construction and operational data. For EPC firms, BIM enhances collaboration among geographically dispersed teams by providing a common data environment (CDE) where project information can be updated and accessed in real time. Rinchen (2024) notes that BIM adoption between 2013–2023 has expanded beyond design coordination to include procurement planning, clash detection and lifecycle management. By integrating BIM with other digital tools such as IoT and digital twins, EPC firms can improve accuracy in design, reduce rework during construction and enhance long term asset performance. However, successful BIM adoption requires strong leadership support and clear standards for data governance, otherwise firms risk underutilizing the technology or creating silos of incomplete information. BIM is therefore an important component of digital transformation in EPC firms, bridging technical efficiency and collaborative workflows.

Table 1. Key Digital Technologies in EPC Project Management

Technology	Main Applications in EPC Projects	Benefits for Project Execution	Key Challenges/Barriers	References
Artificial Intelligence (AI)	Predictive analytics, demand forecasting, risk detection, scheduling optimization	Improves accuracy, reduces delays, enhances decision making	Requires quality data; resistance to automated decision making	Chen (2025); Jing & Fan (2024)
Blockchain	Procurement, contract management, traceability	Transparency, fraud reduction, accountability in supply chains	Technical complexity; lack of industry-wide standards	Jing & Fan (2024); IEEE Xplore (2023)
Automation/ RPA	Document processing, reporting, compliance monitoring	Reduces administrative workload; speeds up processes	Fear of job displacement; need for upskilling	Musaigwa & Swanepoel (2025); HBR (2023)
IoT (Internet of Things)	Real-time equipment/material tracking, monitoring	Improves visibility, predictive maintenance, reduces downtime	Cybersecurity risks; fragmented integration	Sundarakani (2024); Reis (2023)
Digital Twins	Virtual models of assets/projects	Enables scenario modeling, supports resilience, lifecycle management	High implementation cost; data integration challenges	Sundarakani (2024)
Building Information Modelling (BIM)	Shared 3D modeling, design coordination, lifecycle management	Enhances collaboration, reduces rework, improves accuracy	Requires governance standards; cultural resistance to adoption	Rinchen (2024)

The summary in **Table 1.** consolidates insights from recent literature on the key digital technologies most relevant to EPC project management. While each tool offers distinct benefits, their successful adoption depends not only on technical capabilities but also on organizational readiness. AI and blockchain provide transparency and predictive insights, IoT and digital twins strengthen resilience and monitoring, while BIM improves collaboration and lifecycle management. However, barriers such as data integration, cybersecurity risks, lack of governance standards and organizational resistance highlight that technology alone is insufficient. This reinforces the need for complementary organizational enablers, which are discussed in **Section 2.5.**

4.4 Supply Chain Digitalization in EPC Projects

The EPC supply chain is both a critical success factor and a major source of risk. Procurement delays, supplier disputes and lack of visibility across tiers often translate directly into cost overruns and schedule slippage. Literature consistently identifies supply chain transformation as the most impactful area for digitalization in EPC contexts.

Musaigwa and Swanepoel (2025) highlight that AI, blockchain and IoT collectively enhance visibility, agility and accountability across supply chains. Jing and Fan (2024) provide evidence that supply chain digitalization improves integration and reduces procurement risks. Sundarakani (2024) adds a sustainability perspective, showing that digital supply chains support not only efficiency but also resilience in times of global disruption.

Despite these advances, supply chain digitalization remains a challenge for many EPC firms due to cultural resistance, data sharing reluctance and lack of leadership commitment. This reinforces the need to view digital supply chains not as purely technical systems but as sociotechnical ecosystems where technology and human collaboration interact.

4.4.1 Sustainability and Resilience in Digital Supply Chains

An increasingly important dimension of supply chain digitalization is its contribution to sustainability and resilience. Sundarakani (2024) argues that EPC supply chains face dual pressures: ensuring efficiency while simultaneously meeting

sustainability targets and adapting to global disruptions. Digital technologies provide mechanisms for achieving both. For example, blockchain can trace sustainable sourcing of materials, AI can optimize logistics to reduce carbon emissions and IoT sensors can monitor energy use and waste on project sites. Harvard Business Review (2023) highlights that EPC firms embedding sustainability into digital supply chains are more competitive in securing contracts, as clients and governments increasingly demand ESG compliance.

Resilience has also become a priority in the wake of COVID-19, geopolitical instability and climate-driven risks. Digital twins and AI driven scenario modeling allow EPC firms to anticipate disruptions and develop contingency plans. By integrating sustainability and resilience into digital transformation strategies, EPC firms can not only improve execution efficiency but also strengthen their social license to operate in a rapidly changing global environment.

4.5 Organizational Enablers of Digital Transformation

4.5.1 Leadership Commitment

Leadership plays a pivotal role in driving digital initiatives. Reis (2023) shows that projects with strong executive sponsorship achieve higher adoption rates and stronger performance outcomes. Chen (2025) highlights that governance structures and leadership commitment are decisive in determining whether digital transformation remains at the pilot stage or scales across an organization. In EPC firms, where projects involve high risk and complexity, leadership is essential to aligning digital initiatives with strategic objectives.

4.5.2 Cross cultural Collaboration

Multinational EPC projects involve culturally diverse teams that often operate across time zones and organizational boundaries. Differences in communication, decision-making and trust can become barriers to effective digital adoption. Jing and Fan (2024) emphasize that cross cultural misalignment undermines technology adoption in supply chains. The PMBOK (7th edition) also highlights stakeholder and cultural engagement as essential components of successful project execution. Therefore, digital transformation strategies must proactively address cultural barriers ensuring that tools are designed and implemented in ways that support inclusivity and shared understanding.

4.5.3 Change Management

Change management has emerged as one of the most critical organizational enablers of digital transformation. While technologies such as AI, blockchain and BIM promise efficiency and transparency, their implementation often requires significant alterations in workflows, roles and even organizational culture. Without structured approaches to managing this transition, many digital initiatives in project-based industries fail to scale beyond initial pilots (Reis, 2023).

In the context of EPC firms, change management is particularly important due to the complexity and risk profile of large scale, cross border projects. Project teams are often accustomed to established routines and the introduction of digital tools can be perceived as disruptive, generating resistance or mistrust. This resistance may manifest in low adoption rates, parallel reliance on outdated systems or even open conflict between management and operational staff. To mitigate these risks, change management practices must be integrated into the overall transformation strategy.

Kotter's (1996) widely cited eight step model for leading change provides a useful framework for EPC digital initiatives. Its principles such as creating a sense of urgency, forming guiding coalitions and developing and communicating a clear vision are directly applicable to multinational project environments. For example, EPC firms can enhance adoption by involving project managers, engineers and procurement specialists early in the decision-making process, thus creating ownership and reducing resistance. Chen (2025) emphasizes that firms that embed structured change management processes achieve significantly higher rates of adoption and sustained use of digital platforms compared to those that treat transformation as a purely technical implementation.

Furthermore, change management in multinational EPC projects must account for cultural diversity and organizational hierarchies. Communication strategies, training programs and incentive structures should be adapted to ensure that digital adoption is inclusive across different regions and team structures. By embedding change management as a formal enabler of digital transformation, EPC firms can bridge the gap between technological potential and organizational reality, ensuring that investments in digital tools translate into improved project execution.

In summary, organizational enablers form the foundation upon which digital transformation initiatives in EPC firms succeed or fail.

- Leadership commitment provides the strategic direction and resources required for adoption;
- Cross cultural collaboration ensures that multinational teams can work effectively across borders;
- Structured change management processes reduce resistance and align stakeholders around shared objectives; and

- Continuous investment in digital skills equips employees to use advanced tools productively.

These enablers are not isolated, but interdependent: effective leadership must be supported by training, cultural alignment and change management practices. Together, they represent the human and organizational dimension of digital transformation that complements technological innovation. Without these enablers, even the most advanced digital tools risk remaining underutilized, fragmented or resisted by project teams. Conversely, when technological and organizational enablers are integrated, EPC firms are better positioned to achieve execution efficiency, transparency and resilience in global projects.

4.5.4 Digital Skills and Training

Another essential enabler of digital transformation in EPC firms is the development of digital skills and the establishment of continuous training programs. Advanced technologies such as BIM, IoT platforms, blockchain enabled procurement systems and AI driven analytics require technical competencies that go beyond traditional project management skills. Without adequate training, project teams often perceive these tools as disruptive, complex or irrelevant to their day-to-day responsibilities, leading to limited adoption and reduced organizational return on investment (Musaigwa & Swanepoel, 2025).

Digital transformation literature increasingly highlights upskilling and reskilling as strategic imperatives for project based organizations. Harvard Business Review (2024) reports that companies investing systematically in digital training achieve faster adoption of platforms and higher productivity outcomes compared to firms with ad hoc learning initiatives. In EPC contexts, where project success depends on collaboration across engineering, procurement and construction functions, digital readiness at all levels of the organization becomes a decisive factor in whether technologies deliver measurable impact.

Cross cultural dimensions further complicate the challenge of skills development in multinational EPC firms. Teams spread across regions often demonstrate uneven levels of digital literacy, which can exacerbate resistance or create dependencies on a limited number of “digital champions.” Jing and Fan (2024) argue that skill imbalances across global teams frequently delay technology integration, as less digitally mature groups require more time to adapt. Addressing this challenge requires structured, inclusive training programs that accommodate different cultural learning styles, communication norms and baseline competencies.

Moreover, digital skills development must be treated as an ongoing process rather than a onetime intervention. Technologies evolve rapidly and firms that fail to institutionalize continuous learning risk obsolescence. Chen (2025) highlights that EPC firms that embed digital training into professional development frameworks supported by leadership commitment and adequate resources which report greater resilience and innovation capability. By prioritizing digital skills and training, EPC organizations ensure that technological adoption is supported by human capacity, thereby enabling sustained transformation and long-term project execution success.

4.5.5 Communication & Transparency

Effective communication and transparency are essential for successful digital transformation, particularly in complex and multinational EPC environments. Transparent information sharing enables project teams to coordinate effectively, reduces ambiguity and builds trust among stakeholders. The PMBOK (7th edition) identifies communication as a critical component of stakeholder management, emphasizing that clarity and openness are vital for maintaining alignment between project goals and execution practices.

In EPC projects, digital platforms such as cloud collaboration tools, BIM dashboards and AI-driven reporting systems provide real-time visibility of project progress. However, their effectiveness depends on leadership and governance support to promote a culture of openness. Jing and Fan (2024) highlight that communication gaps often undermine technology adoption, especially when teams operate across time zones and cultural contexts. Establishing structured communication channels and transparent reporting frameworks ensures that decisions are made based on accurate, shared information.

Ultimately, communication and transparency act as connective enablers, linking leadership, collaboration and change management. When embedded into the organization's digital ecosystem, they enhance accountability, reduce misunderstandings and improve overall project resilience.

4.5.6 Governance & Policy Support

Governance and policy support form the structural backbone of digital transformation in EPC firms. Governance ensures that digital initiatives align with organizational goals, comply with regulations and uphold ethical and data protection standards. Reis (2023) argues that governance is a decisive factor in scaling transformation projects beyond the pilot phase, while Chen (2025) adds that well-defined governance frameworks reduce risk and ensure accountability in project delivery.

In EPC projects, governance includes setting clear roles for data management, defining decision hierarchies, and establishing policies for interoperability and cybersecurity. International standards such as ISO 19650 (for BIM) and GDPR (for data protection) illustrate how governance connects digital innovation with legal and ethical compliance. Harvard Business Review (2024) further emphasizes that governance maturity directly correlates with digital success: firms with transparent policy frameworks achieve higher adoption rates and reduced operational risks.

Governance and policy support also ensure sustainability and continuity in digital transformation. They provide accountability mechanisms, performance metrics and evaluation structures that maintain focus even after initial implementation. In EPC contexts, where projects are large, multidisciplinary and globally distributed, strong governance ensures consistency and credibility across all digital operations.

Collectively, the six organizational enablers are leadership commitment, cross cultural collaboration, change management, digital skills and training, communication and transparency and governance and policy support which form the foundation for effective digital transformation in EPC firms.

These enablers are interdependent: leadership drives vision, collaboration promotes inclusivity, communication ensures transparency, change management reduces resistance, skills foster competence and governance sustains accountability. When integrated, they create a systemic foundation for innovation, efficiency and sustainable project execution.

4.6 Supporting Theoretical Models

4.6.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM), originally developed by Davis (1989), has been one of the most widely used frameworks to explain technology adoption at the individual level. The model identifies two core determinants: perceived usefulness (PU) or the extent to which a user believes a system enhances job performance and perceived ease of use (PEOU), or the degree to which a system is perceived as free of effort. These factors together shape the behavioral intention to use technology, which subsequently influences actual usage behavior.

In EPC project environments, TAM is highly relevant because adoption of new digital tools such as AI enabled decision platforms, blockchain based procurement systems and BIM which often depends on how engineers, managers and procurement staff perceive their value. If employees see these systems as useful in reducing delays or simplifying tasks, adoption is more likely. Conversely, if tools are seen as complicated or irrelevant to daily work, resistance occurs even when leadership mandates their use.

Recent extensions of TAM have adapted the framework to digital transformation contexts. Chen (2025) highlights that organizational support and leadership communication are additional drivers of adoption in project settings. In multinational EPC firms, this is compounded by cross cultural variation in risk perception, openness to change and digital literacy (Jing & Fan, 2024). Thus, in this thesis, TAM provides a useful lens to examine how individual perceptions of usefulness and usability interact with broader organizational enablers to determine adoption success.

4.6.2 Systems Thinking

Systems thinking, popularized by Senge (1990), emphasizes that organizations should be understood as complex systems of interrelated components. Rather than analyzing problems in isolation, systems thinking focuses on feedback loops, causal relationships and interdependencies that shape organizational outcomes.

In the context of digital transformation, systems thinking is crucial because EPC projects involve multiple interconnected subsystems: engineering design, procurement, logistics, construction and stakeholder management. Digital tools may improve one subsystem but create unintended consequences in another if not integrated holistically. For example, blockchain adoption in procurement can improve transparency but may slow down processes if supplier systems are not ready to interface with the platform. Similarly, AI based scheduling may generate efficiencies but require data inputs that some project teams cannot provide.

Reis (2023) stresses that a systems perspective is vital for understanding why many digital initiatives fail: organizations often implement tools without considering their impact across interdependent processes. By applying systems thinking, this thesis will map the relationships between technological enablers (AI, blockchain, BIM) and organizational enablers (leadership, culture, skills, change management), showing how they interact as part of a broader sociotechnical ecosystem.

4.6.3 Project Management Body of Knowledge (PMBOK)

The Project Management Body of Knowledge (PMBOK), published by the Project Management Institute (PMI), provides a comprehensive framework of processes, tools, and practices for project execution. The 7th edition (2021) shifts emphasis from prescriptive process groups to principle-based project management, making it more adaptable to dynamic and digital contexts.

PMBOK remains highly relevant for EPC firms, which typically manage large, complex projects requiring strict governance of cost, time, scope and risk. Digital transformation can be evaluated effectively within this framework:

- **Scope Management:** AI and BIM tools help refine project scope by providing predictive insights.

- Time and Cost Management: Blockchain and automation improve forecasting and reduce administrative delays.
- Risk Management: Systems thinking enables early identification of digital adoption risks across supply chains.
- Stakeholder and Communication Management: Cross cultural collaboration and digital platforms improve communication and stakeholder alignment.

Recent articles in the *Project Management Journal* (2022–2024) show that PMBOK principles are being applied to digital adoption projects, emphasizing agile approaches, iterative development and greater stakeholder involvement. For EPC projects, aligning digital tools with PMBOK principles ensures that digital transformation is not pursued in isolation but embedded in recognized project management practices.

4.6.4 Integration of Models

While TAM, Systems Thinking and PMBOK each provide valuable insights, their integration offers a more holistic theoretical foundation. TAM explains individual acceptance of digital tools, Systems Thinking highlights organizational interdependencies and PMBOK provides structured processes for managing project execution. Taken together, they reflect the dual nature of digital transformation in EPC: success depends on individual perceptions, organizational dynamics and project governance structures.

This integration underscores the central argument of the thesis: digital transformation in EPC projects cannot be understood purely through technology adoption or process frameworks alone. Instead, it requires an interdisciplinary approach that combines technological, organizational and managerial perspectives to build a comprehensive framework for execution success.

4.7 Summary of Theoretical Framework

This review demonstrates that digital transformation in EPC projects must be understood as a sociotechnical process, requiring the integration of both technological and organizational dimensions. On the technological side, tools such as AI, blockchain, automation, IoT, digital twins and BIM offer opportunities to improve efficiency, transparency and resilience in global supply chains. On the organizational side, leadership commitment, cross cultural collaboration, structured

change management and continuous digital skills development emerge as critical enablers of adoption. The literature emphasizes that without these organizational supports, advanced technologies often remain underutilized or resisted by project teams, undermining their potential impact.

Theoretical models provide additional depth to this framework. TAM explains how user perceptions of usefulness and ease of use shape technology adoption; Systems Thinking highlights interdependencies across organizational subsystems and the unintended consequences of isolated interventions and PMBOK ensures that digital initiatives are aligned with standardized project management practices. Taken together, these models reinforce the need for a multi level approach that considers individual, organizational and project governance perspectives simultaneously.

The synthesis of these insights provides a robust foundation for the thesis. By combining technological enablers, organizational enablers, and theoretical models, this research will contribute to the development of a strategic digital transformation framework tailored to multinational EPC projects. This framework will directly address the research questions by examining (1) how digital tools enhance supply chain efficiency and transparency and (2) how leadership and cross cultural collaboration influence adoption.

The next chapter builds on this theoretical foundation to present the research methodology, outlining the design, data collection and analysis methods that will be used to investigate these issues in practice.

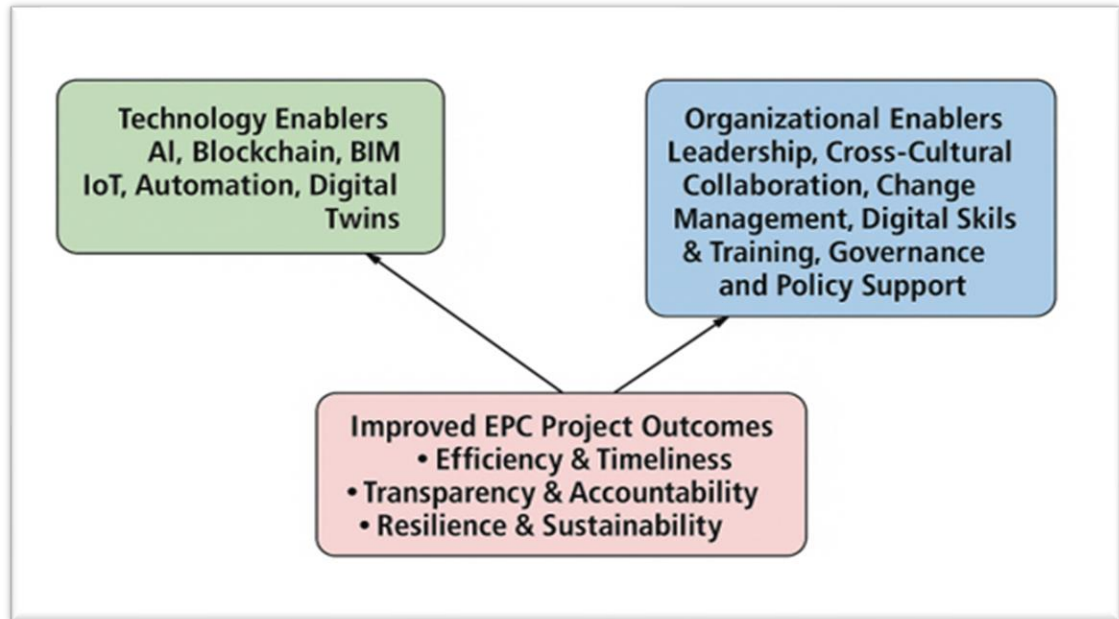


Figure 1. Conceptual Framework of Digital Transformation

Technological enablers such as AI, blockchain, BIM, IoT, automation and digital twins drive operational capability, while organizational enablers leadership, cross cultural collaboration, change management, digital skills and training, communication and transparency and governance and policy support which ensure adoption and sustainability.

Together, these enablers lead to improved EPC project outcomes in terms of efficiency, transparency, resilience and sustainability.

The framework highlights that project outcomes cannot be explained by technology alone. Instead, improvements in efficiency, transparency, accountability and resilience are the result of a dynamic interaction between digital tools and organizational readiness. By integrating both dimensions, this framework directly supports the research objectives and questions of the thesis. It also provides the basis for the methodological choices discussed in Chapter 3, guiding how data will be collected and analyzed to test and refine this conceptual model in the context of multinational EPC firms.

5 METHODOLOGY

5.1 Research Philosophy

The choice of research philosophy forms the foundation for methodological decisions in any study. Research philosophies in social sciences are typically grouped into three main categories: positivism, interpretivism and pragmatism (Saunders, Lewis & Thornhill, 2019). Each carries distinct assumptions about reality, knowledge and the role of the researcher.

Positivism is rooted in the natural sciences and assumes that reality is objective, measurable and independent of human perception. Positivist research often relies on quantitative data, statistical testing and hypothesis validation. While this approach provides generalizable results, it is limited in its ability to capture the nuanced and context-specific dynamics of organizational change and cultural interactions within EPC firms.

Interpretivism, by contrast, emphasizes the socially constructed nature of reality. It focuses on understanding phenomena through the subjective meanings and experiences of participants. Interpretivist approaches are particularly useful in exploring leadership, organizational culture and cross cultural collaboration. However, interpretivism alone risks producing findings that are highly contextual but lack broader generalizability.

Given the dual nature of this research addressing both technological adoption (which benefits from measurable data) and organizational enablers (which require interpretive insights), this study adopts a pragmatic research philosophy. Pragmatism rejects the rigid dichotomy between positivism and interpretivism and instead prioritizes the use of methods that best answer the research questions (Creswell, 2014). Pragmatists view knowledge as provisional and action-oriented, focusing on what works in practice.

This philosophy is especially appropriate for studying digital transformation in EPC firms, where real world execution challenges involve both quantifiable patterns (e.g., survey data on technology adoption) and qualitative complexity (e.g., interviews on leadership and cultural issues). By adopting pragmatism, this study is able to integrate quantitative methods (to measure adoption levels and rela-

tionships) with qualitative methods (to capture in-depth experiences and contextual factors). The result is a holistic methodological approach that reflects the sociotechnical nature of digital transformation.

5.2 Research Design

The research design outlines the overall strategy adopted to integrate different elements of this study in a coherent and logical way, ensuring that the research questions are effectively addressed. Since this thesis seeks to investigate both the technological and organizational enablers of digital transformation in EPC firms, it requires a design that balances exploratory depth with measurable evidence.

Research designs are commonly classified as exploratory, descriptive or explanatory (Saunders, Lewis & Thornhill, 2019). An exploratory design is appropriate when the aim is to gain new insights and clarify the nature of a problem, particularly in under researched areas. An explanatory design, on the other hand, seeks to establish causal relationships between variables, often through quantitative hypothesis testing. Descriptive designs lie between these two, focusing on portraying an accurate profile of events or phenomena.

This study primarily adopts an exploratory design because digital transformation in EPC project management remains an emerging field with limited integrated research on technological and organizational dimensions. An exploratory approach allows the researcher to probe the challenges, perceptions and practices of industry professionals without imposing rigid preconceptions. At the same time, the study also incorporates elements of a descriptive design, as it seeks to map the extent of adoption of specific technologies (AI, blockchain, BIM, IoT, automation) and organizational enablers (leadership, culture, skills).

To operationalize this, the study employs a mixed methods strategy, combining both qualitative and quantitative approaches. As Creswell and Plano Clark (2018) explain, mixed methods research is particularly valuable when neither qualitative nor quantitative approaches alone are sufficient to capture the complexity of a problem. In this thesis:

1. The qualitative strand (semi structured interviews, expert panels, case study) provides rich insights into how digital transformation is perceived, implemented, and enabled within EPC contexts.

2. The quantitative strand (survey) complements this by generating measurable evidence on adoption levels, user perceptions and organizational support factors.

The mixed methods approach also ensures triangulation, where findings from one method validate and reinforce those from another. More importantly, it allows for complementarity (where qualitative findings explain quantitative patterns) and development (where results from one strand inform the design of the other) (Greene, Caracelli & Graham, 1989).

This research design is therefore well suited to address the dual research questions of the study: (1) the impact of digital technologies on supply chain efficiency and transparency and (2) the role of leadership and cross cultural collaboration in enabling adoption. By integrating exploratory, descriptive and mixed methods elements, the design reflects both the novelty and the complexity of digital transformation in multinational EPC firms.

5.3 Data Collection Methods

The quality of any research project is strongly influenced by the appropriateness of its data collection methods. Since this study applies a mixed methods design, both primary and secondary data are collected to capture the complexity of digital transformation in EPC project management.

5.3.1 Primary Data

Semi Structured Interviews

Semi-structured interviews are employed to explore the lived experiences and perceptions of professionals working in EPC firms. This method allows for flexibility in probing specific themes while maintaining consistency across participants. Purposive sampling is applied to select 10 participants, including project managers, supply chain heads and IT leaders from multinational EPC firms. The sample ensures diversity across functions that are directly affected by digital transformation.

The interview guide is designed around four thematic areas:

1. Execution challenges in multinational EPC projects.
2. Experiences with digital tools such as AI, blockchain, BIM and IoT.

3. Perceptions of leadership support and organizational readiness.
4. Issues related to cross cultural collaboration and team dynamics.

Each interview is expected to last 30-45 minutes and will be recorded (with consent) for transcription. The guide includes 8-10 open ended questions, such as:

- Can you describe a project where digital tools either improved or failed to improve execution efficiency?
- How do leadership decisions within your firm influence the adoption of digital tools?
- What role do cultural differences play when working with international teams using shared digital platforms?

Expert Panel Discussions

To validate and contextualize findings, an expert panel of 3-4 professionals specializing in AI, blockchain and digital project management will be convened. The panel will critically review preliminary findings from interviews and surveys to ensure alignment with industry realities.

Survey

A structured survey is distributed to 45-50 EPC professionals using snowball sampling, where initial participants recommend further respondents. The survey is divided into four sections:

1. Demographics (role, years of experience, firm type, geographical focus).
2. Technology adoption (extent of AI, blockchain, BIM and IoT usage).
3. Organizational enablers (leadership support, cross cultural collaboration, training).
4. Perceptions of impact (supply chain transparency, project efficiency, resilience).

The survey employs 5 point Likert scales (1 = strongly disagree, 5 = strongly agree). Example items include:

- The digital tools adopted in my firm significantly improve supply chain transparency.
- Leadership commitment is a decisive factor in whether digital tools are adopted in my projects.

- Cross cultural collaboration challenges reduce the effectiveness of digital platforms in our teams.

Survey constructs are explicitly aligned with the Technology Acceptance Model (TAM) (usefulness, ease of use) and extended with constructs related to leadership and cultural enablers, as suggested in recent studies (Chen, 2025; Jing & Fan, 2024).

Case Study- CCC Engineering Ltd. (CCCE)

The case study focuses on CCC Engineering Ltd., a multinational EPC firm operating across Europe, Asia and the Middle East. CCCE was chosen because it represents a typical large scale EPC company actively experimenting with digital solutions but facing persistent execution challenges. The case allows for in depth analysis of how digital transformation strategies are implemented in practice, providing contextual richness that complements the broader survey and interview findings.

5.3.2 Secondary Data

Secondary data is drawn from peer reviewed journals such as *Harvard Business Review*, *Project Management Journal* and *IEEE Xplore Digital Library*, as well as industry benchmark reports from leading EPC firms including Siemens, Fluor and Bechtel. These sources provide comparative insights, highlight best practices and help triangulate primary data findings. Secondary data also ensures that the research is grounded in both academic theory and industry practice.

5.4 Data Analysis

Given the mixed methods design of this study, data analysis combines qualitative and quantitative techniques to ensure that both rich contextual insights and measurable evidence are obtained. This integration strengthens the validity of the findings and allows for a more comprehensive understanding of digital transformation in EPC project management.

5.4.1 Qualitative Analysis

Interview transcripts and expert panel notes will be analyzed using thematic coding. Both deductive coding (based on predefined themes from

the literature, such as leadership commitment, cultural collaboration and technology adoption) and inductive coding (emerging patterns from the data itself) will be applied. This hybrid approach ensures that the analysis is grounded in established theory while remaining open to new insights from practice (Braun & Clarke, 2019).

The coding process will follow three stages:

1. Initial coding to highlight recurring words, phrases and patterns.
2. Axial coding to group codes into broader categories such as execution challenges, enablers of adoption and barriers to implementation.
3. Selective coding to connect categories to the overarching research questions.

NVivo or a similar qualitative analysis software will be used to manage transcripts and maintain consistency. Cross case analysis will then be conducted by comparing findings from CCCE with other EPC firms represented in the interviews, highlighting both commonalities and firm specific variations.

5.4.2 Quantitative Analysis

Survey data will be analyzed using descriptive and inferential statistics.

- Descriptive statistics (frequencies, means and percentages) will summarize adoption levels of AI, blockchain, BIM and IoT as well as perceptions of leadership support and cultural collaboration.
- Inferential statistics will include correlation and regression analysis to test relationships between constructs. For example:
 - Correlation between *leadership support* and *perceived usefulness of digital tools*.
 - Regression analysis examining whether *cross cultural collaboration* predicts *supply chain transparency outcomes*.

Open ended survey responses will undergo machine learning based sentiment analysis. This technique will classify responses as positive, neutral or negative toward digital transformation initiatives, providing additional insights into leadership and cultural perceptions. Sentiment analysis will complement the thematic coding from interviews by highlighting patterns across a larger sample.

5.4.3 Systems Thinking Approach

In line with the study's theoretical framework, findings will also be analyzed through a systems thinking lens. This involves mapping causal relationships between technological enablers (e.g., AI, blockchain, BIM), organizational enablers (e.g., leadership, change management), and project execution outcomes (e.g., efficiency, transparency, resilience).

For example, a causal loop diagram will illustrate how strong leadership commitment increases adoption of blockchain based procurement systems, which in turn enhances supply chain transparency and reduces execution delays. Feedback loops will also be identified, such as how improved project outcomes reinforce leadership confidence in digital initiatives, creating a virtuous cycle of adoption.

This systems oriented analysis enables the study to go beyond isolated cause-effect relationships, offering a holistic understanding of how digital and organizational enablers interact in multinational EPC projects.

5.5 Validation and Reliability

Ensuring the validity and reliability of findings is essential for producing trustworthy research. In mixed methods studies, these concepts must be addressed across both qualitative and quantitative strands. This study applies multiple strategies to strengthen construct validity, internal validity, external validity and reliability.

Construct Validity

Construct validity concerns whether the data collection tools accurately measure the concepts under investigation. In this thesis, interview questions and survey items are directly aligned with constructs identified in the literature, such as *perceived usefulness* and *ease of use* (TAM), *leadership commitment* and *cross cultural collaboration*. Survey items are formulated as Likert scale statements adapted from validated instruments in prior studies (Chen, 2025; Jing & Fan, 2024). To further enhance construct validity, the survey instrument will be piloted with 2-3 EPC professionals before full distribution, ensuring clarity and relevance.

Internal Validity

Internal validity refers to the credibility of causal interpretations drawn from the study. Since the research is primarily exploratory, causal claims will be made cautiously. However, triangulation of data sources (interviews, surveys, case study, secondary literature) will strengthen internal validity by showing consistency across methods. Additionally, qualitative coding will follow transparent procedures, with clear audit trails documenting how themes were derived.

External Validity

External validity concerns the generalizability of findings. While qualitative findings from interviews and the CCCE case study may be context specific, survey results provide broader insights into adoption trends across multiple EPC firms. Although the relatively small sample size may limit statistical generalization, the study emphasizes analytical generalization (Yin, 2018), where findings are generalized to theoretical propositions rather than to populations.

Reliability

Reliability refers to the consistency of research procedures. To ensure reliability:

- Interview transcripts will be coded systematically using NVivo software.
- Coding protocols and category definitions will be documented so that another researcher could replicate the process.
- Survey administration will follow standardized procedures, with the same instructions given to all participants.
- Data will be stored securely and consistently handled in accordance with GDPR regulations.

Triangulation

Triangulation further enhances both validity and reliability. This study applies:

- Data triangulation (interviews, surveys, case study, secondary data).
- Method triangulation (qualitative + quantitative approaches).
- Investigator triangulation through expert panel reviews of preliminary findings.

By combining these strategies, the study ensures that its findings are both credible and dependable. Together, they strengthen the robustness of the digital transformation framework that will be developed in later chapters.

To further strengthen validity and reliability, the study incorporated consistent procedures across all methods and cross-checked findings between interviews, survey data and the case study. The same interview guide was used for all participants and survey items were adapted from validated instruments, ensuring construct accuracy. Themes identified in interviews were corroborated by quantitative correlations (e.g., leadership and communication showing the strongest relationships), which increases internal validity. Reliability was supported by clear coding protocols and standardized data collection procedures that another researcher could replicate. Because findings converged across multiple sources, the results can be considered credible and dependable within the scope of this exploratory mixed-methods study.

5.6 Ethical Considerations

This study is conducted in accordance with the ethical research guidelines of Tampere University of Applied Sciences (TAMK) and the principles of responsible academic inquiry. Given the sensitivity of organizational information and the professional roles of participants, ethical considerations are prioritized throughout the research process.

Informed Consent

All participants will be fully informed about the purpose, scope and procedures of the study before taking part. Interviewees and survey respondents will receive an information sheet outlining the research aims, the voluntary nature of participation and how their data will be used. Written or digital consent will be obtained prior to data collection.

Confidentiality and Anonymity

Confidentiality is guaranteed by anonymizing all participant responses. Names, company identifiers and other personal data will not appear in transcripts, reports or the final thesis. Instead, responses will be coded (e.g., Interviewee A, EPC Project Manager) to protect identities. This ensures that individuals and firms cannot be identified in the published results.

Data Protection and GDPR Compliance

All data will be handled in compliance with the General Data Protection Regulation (GDPR). Interview recordings and transcripts will be stored on encrypted,

password protected drives. Access will be restricted to the researcher only. Data will be retained securely for a period of five years after thesis submission, after which it will be permanently deleted. Survey data will be collected through a GDPR compliant online platform to ensure additional protection.

Right to Withdraw

Participants will have the right to withdraw from the study at any point without providing justification. If withdrawal occurs, their data will be deleted immediately from the dataset to ensure respect for autonomy.

Bias Minimization

The research design incorporates participants from diverse cultural and organizational backgrounds to reduce bias and ensure balanced perspectives. Reflexivity will be maintained throughout the study, with the researcher acknowledging potential influences of personal experience as a former project manager in EPC contexts. Expert panels will also serve as a safeguard against interpretive bias by validating emerging findings.

By embedding these ethical safeguards into the research process, this study ensures that it respects participants, protects sensitive organizational information and upholds the standards of ethical and responsible academic research.

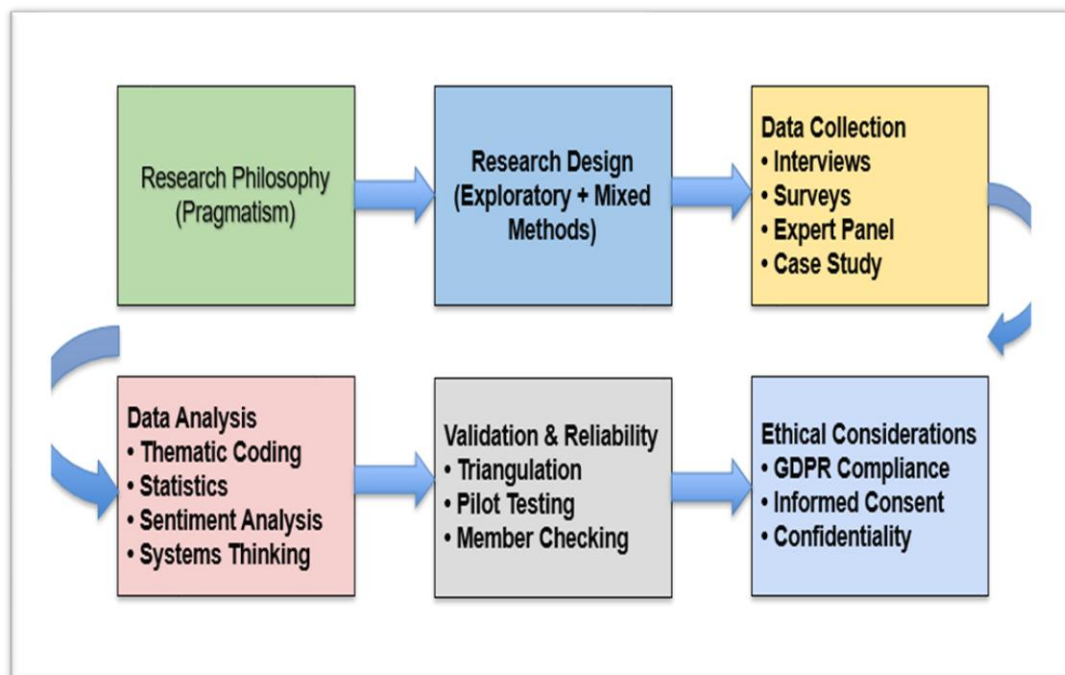


Figure 2. Methodology Flow Diagram

The figure presents the pragmatic research philosophy and exploratory mixed methods design guiding this study. It shows the progression from research design through data collection and analysis to validation and ethical considerations, ensuring reliability, transparency and compliance with responsible academic standards.

5.7 Summary of Methodology

This chapter has outlined the methodological approach used to investigate digital transformation in multinational EPC firms. A pragmatic research philosophy was adopted to combine the strengths of both qualitative and quantitative methods, recognizing the sociotechnical nature of the research problem. The research design was framed as exploratory with descriptive elements, employing a mixed methods strategy to capture both depth and breadth of insights.

Primary data collection combines semi structured interviews, expert panels, surveys and a case study of CCC Engineering Ltd., while secondary data from peer reviewed journals and industry reports ensures grounding in existing knowledge and benchmarking against best practices. Data analysis integrates thematic coding, cross case comparison, statistical analysis, sentiment analysis and a systems thinking approach to map interdependencies between technological and organizational enablers.

Validity and reliability are enhanced through careful alignment of constructs with established theories (TAM, Systems Thinking, PMBOK), triangulation of data sources and methods, pilot testing, and peer debriefing. Ethical safeguards, including informed consent, GDPR compliant data protection and participants right to withdraw ensure that the study adheres to the highest standards of responsible research.

Together, these methodological choices create a robust foundation for the empirical investigation. The following chapter presents the results of the data collection and analysis, offering evidence based insights into how digital tools and organizational enablers shape project execution outcomes in multinational EPC firms.

6 EMPIRICAL FINDINGS

6.1 Introduction to Results

This chapter presents the empirical results derived from both qualitative and quantitative data collection processes designed to address the research objectives and questions defined in Chapter 3. The findings integrate insights from semi structured interviews, a focused case study (CCC Engineering Ltd.) and a structured quantitative survey conducted among professionals in multinational Engineering, Procurement and Construction (EPC) firms across Finland, Germany and Bangladesh.

The central aim of this research was to investigate how digital transformation tools including Artificial Intelligence (AI), Blockchain, Building Information Modeling (BIM), the Internet of Things (IoT) and Automation which contribute to overcoming execution challenges in strategic project management. The study further explored the role of organizational enablers such as leadership commitment, cross cultural collaboration, change management, digital skills, communication transparency and governance support in facilitating digital adoption and operational efficiency.

To achieve methodological triangulation, the data were collected through three complementary phases:

Qualitative Inquiry: Semi structured interviews with project and digital transformation professionals provided in depth understanding of implementation challenges and leadership responses.

Case Study Analysis: The real world example of CCC Engineering Ltd. illustrated practical applications of digital tools and organizational responses within a multinational EPC context.

Quantitative Survey: A Google Forms based survey gathered numerical and perceptual data from 47 respondents, allowing measurement of adoption levels, enabler effectiveness and perceived project impacts across the three geographical regions.

The mixed methods approach enabled both exploratory and confirmatory analysis, providing a balanced interpretation of digital transformation from human, organizational, and technical perspectives. While the qualitative phase emphasized narrative depth and context specific insights, the quantitative phase validated

these observations statistically, revealing measurable relationships between digital tools, enablers and project performance outcomes.

The results are presented in the following order:

- Section 6.2 details the qualitative findings from interviews and the CCC Engineering case study.
- Section 6.3 outlines the quantitative survey results, including descriptive, inferential and sentiment analyses.
- Section 6.4 synthesizes these findings using Systems Thinking to map causal relationships between enablers and project outcomes.
- Finally, Section 6.5 summarizes the key insights emerging from the combined data and prepares the groundwork for the Discussion in Chapter 7.

Overall, this chapter provides a comprehensive evidence base illustrating how strategic digital transformation supported by strong leadership, adaptive change management and transparent communication which enhances project execution capability and organizational resilience in multinational EPC environments.

6.2 Qualitative Findings

6.2.1 Semi structured Interviews

A total of Ten (10) semi structured interviews were conducted with project managers, procurement heads and digital transformation leads from multinational EPC firms in Bangladesh. Thematic analysis identified six recurring organizational enablers influencing digital adoption: leadership commitment, cross cultural collaboration, change management, digital skills, communication and governance.

Key patterns emerged as follows:

Leadership Commitment: Interviewees emphasized that leadership sponsorship and resource allocation were essential for sustaining digital initiatives. One respondent from Bangladesh noted, without senior management's direct involvement, digital initiatives lose momentum after pilot testing.

Cross cultural Collaboration: Respondents from Finland and Germany highlighted cultural differences in risk perception and communication styles as barriers to unified adoption. A project manager stated, While European teams favor structured rollouts, Asian partners often rely on informal coordination.

Change Management: Several participants described resistance among middle managers who feared process disruption. Structured training and communication were cited as effective mitigation strategies.

Digital Skills & Training: Interviewees stressed the need for regular upskilling. “Our engineers master traditional project tools but hesitate with AI based systems,” observed a German IT lead.

Communication & Transparency: The use of digital dashboards and BIM collaboration platforms was praised for improving coordination. However, respondents warned that transparency requires consistent leadership enforcement.

Governance & Policy Support: Participants noted the importance of having clear accountability frameworks for data integrity and cybersecurity.

In summary, the interviews confirmed that leadership, communication and governance act as high level enablers, while training and change management operate as balancing mechanisms supporting digital adoption.

6.2.1.1 Overview of Interviews

To explore the practical realities of digital transformation in the EPC sector and understand how organizations overcome execution challenges, ten semi structured interviews were conducted with senior professionals representing CCC Engineering Ltd. and its partner ecosystem. Participants included project directors, technical leads, systems managers and training heads from both public and private EPC organizations such as PGCB, BPDB, Summit Power, Energypac, Siemens Energy, Confidence Infrastructure, Reverie Power, TBEA Bangladesh and KEC International. The respondents possessed between 10 and 22 years of industry experience, ensuring a mix of strategic, managerial and operational perspectives.

Each interview lasted approximately 45 to 50 minutes and followed a guided conversational format focused on seven analytical dimensions: Leadership Commitment, Cross cultural Collaboration, Change Management, Digital Skills and Train-

ing, Communication and Transparency, Governance and Accountability and Supply chain Integration and Disruption. All interviews were recorded with consent, transcribed and thematically coded using a hybrid deductive inductive approach.

The following table a summary of interview participants, their organizational affiliations and the key findings that emerged. The table illustrates that, while each organization operates within a distinct context, there is a shared pattern: digital transformation initiatives are most effective when leadership vision, human capability and governance discipline converge to mitigate operational and supply chain uncertainties.

Table 2. Overview of Interviews

No.	Organization / Role	Ex- peri- ence	Core Discus- sion Themes	Key Findings (Aligned with Thesis Framework)
1	CCC Engineering Ltd.- Interviewee (20 yrs exp.)	20 yrs	Digital Leader- ship, Govern- ance, Change Management, <i>Supply chain Integration</i>	CCCE's transformation addressed chronic execution inefficiencies by adopting BIM and ERP systems. Leadership visibility and Siemens supported training improved accountability and transparency. Persistent challenges include data integration with joint-venture partners and uneven digital adoption at remote sites, especially in supply chain tracking of imported equipment.
2	Power Grid Company of Bangladesh (PGCB)- Superintend- ing Engineer	16 yrs	Governance & Policy Support, Client Contrac- tor Collabora- tion, Digital Transparency	PGCB's ERP, SCADA and GIS plat- forms standardized communication with EPC partners such as CCCE. Leadership enforced digital govern- ance KPIs. Interoperability gaps and cybersecurity remain concerns. Con- firms that strong client governance accelerates EPC digital maturity.

No.	Organization / Role	Ex- peri- ence	Core Discus- sion Themes	Key Findings (Aligned with Thesis Framework)
3	Energypac Power Systems Ltd.- Head of ICT & Training	15 yrs	Digital Skills & Training, Change Management, Leadership Engagement	Energypac embedded SAP S/4HANA and BIM via a “Digital Academy.” Continuous learning built employee ownership of change. Key challenges: supplier digital immaturity, cybersecurity and aligning training outcomes with measurable performance. Demonstrates that workforce capability sustains transformation.
4	Summit Power Ltd.- Project Director	18 yrs	Communication, Transparency, Integration, <i>Supply chain Coordination</i>	ERP and Primavera improved multi-plant coordination and data sharing. Leadership mandated dashboard use for decision making. Disruptions arise from incompatible digital systems among contractors and vendors, exposing supply chain fragility. Shows that transparent communication enhances responsiveness during disruption.
5	Siemens Energy Bangladesh- Technical Lead	12 yrs	Cross cultural Collaboration, Global Governance, Leadership Influence	Siemens’ BIM 360 and Teamcenter unified international workflows with CCCE. Leadership alignment across borders ensured governance compliance. Main issues: differing digital maturity levels and cyber risk exposure. Illustrates that cross cultural alignment and data security underpin successful multinational collaboration.

6	Confidence Infrastructure Ltd.- Change Management Lead	14 yrs	Change Management, Employee Motivation, Governance	Oracle ERP and Power BI enhanced reporting and analytics. Leadership participation and recognition programs encouraged adoption. Resistance persists among senior staff and manual subcontractors. Reinforces that inclusive change management drives digital confidence.
7	Bangladesh Power Development Board (BPDB)- System Administrator	17 yrs	Governance & Accountability, Leadership Oversight, Public Sector Integration	BPDB's e-GP and PMIS platforms increased transparency in public sector EPCs. Leadership formed a Digital Governance Committee. Challenges: interoperability with utilities, rural connectivity, and cyber preparedness. Demonstrates governance reform as a cornerstone of digital credibility.
8	Reverie Power & Automation Engineering Ltd.- Project Manager	10 yrs	Leadership, Workflow Integration, Governance, <i>Supplier Digitalization</i>	BIM 360 and ERPNext improved coordination and audit readiness. ISO-aligned governance and leadership investment boosted client confidence. Manual supplier processes still disrupt procurement cycles. Reflects CCCE's mid tier replication of leadership driven digitalization.
9	TBEA Bangladesh (Chinese JV)- Project Manager	10 yrs	Cross cultural Leadership, Digital Integration, <i>Import-Logistics Governance</i>	Project Cloud connected Chinese and Bangladeshi teams for transparent coordination. Joint leadership dashboards bridged cultural gaps. Key challenges: port congestion, manual subcontractors and unstable connectivity. Aligns with thesis theme of cross cultural digital execution.

10	KEC International Ltd. (Bangladesh Ops.)- Project Director	22 yrs	<i>Supply chain</i> <i>Digitalization,</i> Leadership Commitment, Governance In- tegration	SAP ERP and KEC Digital Project Monitoring platform strengthened live supply chain visibility and compliance. Leadership oversight institutionalized predictive analytics. Integration gaps with local partners and data security risks persist. Reinforces that digital supply chain control reduces execution delays.
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Following the overview table, the thematic analysis elaborates on how these dimensions interact to shape the outcomes of digital transformation across CCC Engineering Ltd. and its partner organizations.

6.2.1.2 Thematic Analysis of Interviews

Drawing on all ten interviews, qualitative coding revealed seven interconnected themes that define digital transformation dynamics and execution challenges within CCC Engineering Ltd. and its partner network.

Theme 1 Leadership Commitment and Vision

Leadership consistently emerged as the *core catalyst* for transformation.

In CCCE, Summit Power and KEC, senior executives personally used dashboards and digital review meetings to signal that transformation was strategic, not optional. Participants emphasized that leadership visibility legitimized change and inspired adoption at all organizational levels.

Interpretation: Digital transformation succeeds when leadership commitment moves beyond rhetoric to active participation, converting digital policy into measurable practice.

Theme 2 Cross cultural Collaboration and Alignment

In joint ventures like Siemens & CCCE and TBEA Bangladesh, cultural and procedural differences initially hindered efficiency.

Bilingual training, shared dashboards and unified approval workflows bridged gaps.

Interpretation: Cross cultural digital integration transforms diversity from a barrier into a performance multiplier when systems standardize information flow.

Theme 3 Change Management and Employee Engagement

Almost all firms faced resistance from experienced engineers.

CCCE's *digital champions* and Confidence Infrastructure's *change ambassadors* reduced anxiety and encouraged peer learning. Energypac institutionalized continuous upskilling through its Digital Academy.

Interpretation: Sustained change management and visible recognition convert apprehension into empowerment, ensuring durable adoption.

Theme 4 Digital Skills and Training Capacity

Skill development is the backbone of transformation.

Training programs at CCCE, Energypac and Siemens built a shared digital language across departments and partner organizations. Firms reported tangible productivity gains and morale boosts following structured training interventions.

Interpretation: Competence building transforms technology acceptance into operational excellence, aligning with the thesis model of capability driven maturity.

Theme 5 Communication and Transparency

ERP and BIM integration reduced departmental silos and fostered real time collaboration.

Summit Power's and Reverie Power's dashboards replaced fragmented email reporting, accelerating decisions and accountability.

Interpretation: Transparent data environments enhance coordination, reduce internal politics, and underpin trust a recurring determinant of project success in CCCE's ecosystem.

Theme 6 Governance and Accountability

Public- and private sector participants highlighted that digitization strengthened audit readiness and traceability.

CCCE's SAP approvals, BPDB's e-GP and Siemens' global quality audits exemplify robust digital governance.

Interpretation: Governance is both a control mechanism and a reputation builder; digital audit trails convert compliance into credibility.

Theme 7 Supply Chain Disruption and Integration

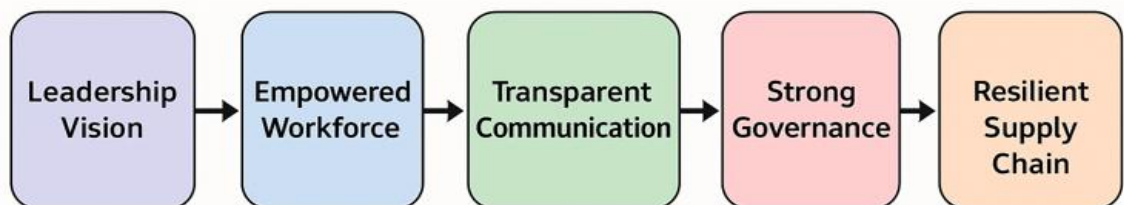
All organizations, especially CCCE, KEC, TBEA and Summit, identified supply chain volatility as the dominant execution challenge.

Material delays, import bottlenecks and vendor digital immaturity disrupt schedules despite advanced planning tools. Firms are mitigating this through predictive analytics, live shipment trackers and supplier training programs.

Interpretation: Supply chain resilience is now a *strategic outcome* of digital transformation. The thesis confirms that ERP driven and BIM driven integration across procurement, logistics and construction is essential to overcoming execution uncertainty.

Cross Theme Synthesis

The thematic interplay reveals a virtuous cycle:



Each enabler reinforces the others, producing measurable improvements in efficiency, accountability and schedule reliability.

However, challenges remain fragmented data among partners, cyber risk exposure and inconsistent vendor participation continue to constrain full digital maturity.

For CCC Engineering Ltd., these findings underscore that digital transformation is a *continuous strategic project*, requiring iterative leadership engagement, investment in people and adaptive governance to sustain competitiveness in Bangladesh's evolving EPC landscape.

6.2.1.3 Cross Theme Interpretation

The seven themes identified through the interviews reveal an interconnected system in which leadership, governance, communication and supply chain integration reinforce one another. Leadership establishes direction and credibility, governance provides structure and accountability, communication ensures transparency and collaboration and digital supply chain integration translates strategic intent into measurable efficiency.

Across all ten interviews, participants emphasized that transformation success depends not only on technological capability but also on organizational culture, the willingness of people to adopt new tools, share information openly and coordinate decisions through data rather than hierarchy. Respondents agreed that execution challenges such as fragmented supplier networks, inconsistent data standards and cultural resistance which remain the most persistent barriers, yet these can be progressively overcome through continuous leadership engagement, targeted digital skills training and cross organizational transparency.

Collectively, the interviews suggest that CCC Engineering Ltd.'s experience reflects a broader pattern within Bangladesh's EPC sector: digital transformation is not a finite project but an evolving strategic capability. As organizations deepen governance maturity and digital integration across supply chains, they move closer to achieving resilient, transparent and globally competitive project management performance.

Together, these themes illustrate that CCC Engineering Ltd. and its partners view digital transformation as a continuous strategic project rather than a onetime ini-

tiative. Leadership engagement and transparent communication underpin a culture that enables change, strengthens governance and mitigates supply chain disruptions.

6.2.1.4 Argumentation Based on Ten Interviews

The ten semi structured interviews provide strong empirical justification for the digital transformation enablers identified in this study. Across all organizations CCCE, PGCB, BPDB, Siemens Energy, Summit Power, Energypac, Confidence Infrastructure, Reverie Power, TBEA Bangladesh and KEC International whereas participants repeatedly emphasized that leadership commitment, transparent communication, governance discipline and supply-chain visibility were decisive factors in overcoming execution challenges. This consistency across diverse organizational contexts strengthens the argument that the enablers identified are not isolated observations but represent systemic patterns within multinational EPC environments.

The interviews collectively support **Research Question 2**, as participants consistently stressed that leadership engagement was the catalyst for adoption. Interviewees from CCCE, Summit Power and KEC International described how top management visibility such as leading dashboard reviews and mandating ERP-based reporting drove behavioral change and built trust in digital tools. This convergence of perspectives reinforces the Systems Thinking **R1 loop (Leadership-Training-Efficiency)**, demonstrating how leadership investment accelerates capability development and improves project performance.

Similarly, cross-cultural collaboration emerged as an important determinant of transformation success, particularly in joint-venture settings involving Chinese, German and Bangladeshi teams. Interviewees from Siemens Energy and TBEA Bangladesh described how bilingual dashboards, shared workflows and transparent communication reduced cultural friction and improved coordination. These findings directly support **R2 (Collaboration-Resilience)**, illustrating how collaboration strengthens resilience and reduces delays.

The interview data also substantiate the balancing role of change management and skills development. Participants from Energypac, Confidence Infrastructure

and CCCE reported that early resistance from senior engineers, vendor immaturity and variations in digital literacy slowed adoption. However, structured training programs, digital champions and peer learning mechanisms reduced resistance and stabilized the transition, validating **B1 (Change-Communication-Adaptation)** as a stabilizing loop.

In addition, interviewees repeatedly highlighted supply-chain disruption imports delays, non-digital subcontractors and weak integration as the most persistent execution barrier. The adoption of ERP-based logistics dashboards and predictive tracking tools at CCCE, KEC and Summit Power demonstrated how digital visibility transformed uncertainty into actionable control. This aligns with the quantitative correlation between supply-chain integration and project efficiency ($r = 0.78$), and reinforces **R3 (Supply-Chain Visibility)** as a strategic enabler.

Overall, the ten interviews provide robust argumentation for the study's conceptual and empirical conclusions. The recurring themes across organizations confirm that digital transformation in EPC firms is not driven by technology alone but by leadership, governance, communication, cultural alignment and supply-chain integration. Their combined effects form a mutually reinforcing system that influences adoption, performance and organizational resilience directly supporting both research questions and validating the systemic framework developed in this thesis.

6.2.2 Case Study- CCC Engineering Ltd.

6.2.2.1 Background of CCCE

CCC Engineering Ltd. (CCCE) was founded in 2000 and is headquartered in Beijing, China. The company specializes in turnkey EPC contracting across the full power chain including generation, transformation, transmission and distribution. With over two decades of experience, CCCE has built a global presence, executing projects in Southeast Asia, South Asia (including Bangladesh), the Middle East and Africa.

In Bangladesh, CCCE has been a key player in the power transmission sector, collaborating with donor funded projects and international consortia. For exam-

ple, a large transmission upgrade project under Power Grid Company of Bangladesh (PGCB) valued at more than USD 1.12 billion was awarded to a consortium including CCCE.

This case study focuses on how CCCE has navigated the digital transformation journey within its Bangladesh operations (and global partner network) to address execution challenges especially in large scale, complex, cross border EPC projects.

6.2.2.2 Digital Transformation Initiatives

CCCE's approach to digital transformation is multi-faceted, encompassing both technological enablers and organizational change. Below are key facets aligned with your thesis enablers:

Leadership Commitment: Senior leadership within CCCE declared digital transformation a strategic objective and elevated it from a purely technical initiative to a core business strategy driver. Visible leadership commitment helped to foster a culture of change across functions.

Digital Skills & Training: CCCE established training programs to upskill engineers in ERP, BIM, IoT dashboards and real time reporting. These programs emphasized that digital tools are enablers for project delivery, not mere IT add-ons.

Governance & Policy Support: The company implemented structured governance mechanisms to monitor digital tool adoption, enforce data integrity and maintain audit compliance (especially important in projects funded by donor agencies).

Communication & Transparency: CCCE transformed its information flow culture by embedding dashboards, centralized data repositories and live progress tracking. Project stakeholders (design, procurement, construction) accessed unified data rather than working in departmental silos.

Cross cultural Collaboration: In Bangladesh, CCCE collaborates with international partners and local firms. Digital systems helped bridge cultural, procedural and language barriers by providing a common platform and standardized workflows across geographies.

Change Management: Recognizing the human side of digitalization, CCCE leveraged change champions, peer training and visible demonstration of benefits to reduce resistance and build trust.

Supply chain Integration & Disruption: Given the heavy reliance on imported equipment and complex logistics, CCCE adopted digital supply chain tracking and integrated data flows between global suppliers, logistic partners and local sites. This enabled improved transparency, quicker decision-making and reduced execution delays.

6.2.2.3 Execution Challenges & Solutions

CCCE's digital transformation journey has been marked by several execution challenges, many of which mirror your thesis's focus on overcoming project management barriers in multinational settings.

Challenge: Supply chain delays due to complex import logistics, vendor immaturity and non-digital workflows.

Solution: CCCE introduced a digital supply chain tracking dashboard linked to procurement and site planning modules. Leadership oversight of the dashboard enabled early detection of import delays and allowed site managers to adjust schedules proactively.

Challenge: Fragmented coordination across design, procurement and construction teams.

Solution: Implementing a unified ERP/BIM system enabled all stakeholders to work on the same version of truth; design changes, procurement updates and site progress were visible in real time.

Challenge: Cultural and digital maturity gaps between head-office, local operations and partners.

Solution: Bilingual training, joint workshops and co-located teams helped build a unified digital mindset. Leadership from both global and local units engaged in review meetings, which strengthened cross cultural trust and alignment.

Challenge: Ensuring governance and audit readiness in donor-funded projects.

Solution: CCCE developed digital audit trail systems with timestamped approvals and e signatures, enabling traceability and transparency that met donor standards. Auditors reported improved documentation quality and faster review cycles.

6.2.2.4 Lessons Learned- Implications for Strategic Project Management

From CCCE's experience, several lessons emerge that directly support your thesis's proposition:

1. Digital transformation is not about tools alone, it requires leadership commitment, training, governance and full organizational alignment.
2. Supply chain resilience is a strategic outcome of integrating digital tools into procurement and logistics workflows, particularly in complex EPC environments in emerging economies.
3. Cross organizational transparency and communication are critical to overcoming execution delays and coordination failure, both internal and in partner networks.
4. Change management and cultural adaptation must be explicitly managed; technology alone cannot deliver performance without human buy-in and capability building.
5. Governance mechanisms embedded in digital systems strengthen accountability, auditability and client confidence, vital in international EPC contracts and donor funded projects.

In summary, CCCE demonstrates how multinational EPC firms operating in Bangladesh and similar markets can leverage digital transformation to enhance efficiency, transparency, resilience and strategic project delivery capacity. For your thesis, this case study serves as a rich empirical anchor for discussing how the six organizational enablers (leadership, collaboration, change management, digital skills & training, communication & transparency, governance) interact with digital tools to overcome execution challenges in strategic project management.

The convergence of CCC Engineering's experience and the wider industry perspectives underscores that digital transformation is an integrated managerial

challenge. The following section synthesizes these insights to outline the overall qualitative conclusions.

6.2.2.5 Comparative Insights & Integration with Interview Findings

The case study of CCC Engineering Ltd. complements and reinforces the qualitative findings from the ten semi structured interviews. Both datasets collectively demonstrate that digital transformation within Bangladesh's EPC environment is driven by leadership commitment, strengthened by governance and communication and ultimately tested by supply chain resilience.

Alignment of CCCE's Experience with Interview Themes

1. Leadership Commitment

The CCCE case study confirms what interviewees from Summit Power, Energypac and KEC International highlighted visible and sustained leadership engagement is the cornerstone of successful transformation. CCCE's senior management did not treat digitalization as an IT project but as a business strategy initiative. Their decision to personally lead dashboard reviews and insist on ERP based reporting mirrors the leadership behaviors identified as critical in multiple interviews.

2. Change Management and Workforce Adaptation

Both the case and the interviews reveal that employee resistance was one of the earliest obstacles. CCCE's deployment of "digital champions" at project sites parallels the "change ambassadors" and "digital mentors" described in Confidence Infrastructure and Energypac. These findings reinforce the argument that transformation depends more on people readiness than on software capability.

3. Digital Skills and Continuous Learning

Training initiatives such as CCCE's collaboration with Siemens Energy for BIM and ERP training reflect the same workforce development emphasis found across the interview data. The interviews confirm that learning platforms and skill building programs like Energypac's Digital Academy which increase

employee confidence, reduce process errors and improve performance metrics.

4. Communication and Transparency

Interviewees across organizations agreed that communication silos and information delays previously undermined project performance. CCCE's centralized dashboards and real-time progress updates provided a single version of truth that eliminated interdepartmental misalignment. This finding corresponds with evidence from Summit Power and Reverie Power, where ERP and BIM tools enhanced cross departmental coordination and decision speed.

5. Governance and Accountability

The case study demonstrates that CCCE's digital audit trail complete with e-signatures and timestamps which met international compliance standards and improved donor confidence. This echoes the experiences of BPDB and Siemens Energy, where digital documentation strengthened accountability and reduced audit times. Governance thus emerges as the key institutional enabler bridging technology and credibility.

6. Cross cultural Collaboration

CCCE's partnerships with Chinese and European suppliers required standardized reporting and bilingual training, precisely the issues discussed in the TBEA and Siemens interviews. Digital platforms served as neutral meeting grounds, harmonizing differing work cultures and expectations. The convergence of findings underscores that cross cultural collaboration is both a challenge and an opportunity for learning in multinational EPC ecosystems.

7. Supply chain Integration and Disruption

Perhaps the strongest point of convergence lies in the area of supply chain performance. Both the case study and multiple interviews (CCCE, KEC, Summit, TBEA and Reverie) identify material delays, vendor digital immaturity and weak data synchronization as persistent execution bottlenecks. CCCE's introduction of predictive dashboards and real-time logistics tracking partially mitigated these issues but did not eliminate them. This supports the overall thesis

argument that digital transformation enhances visibility and resilience but requires deeper integration across external supplier networks to fully overcome disruption.

Synthesis of Case and Interview Insights

Collectively, the interviews and case analysis converge on four critical propositions:

1. **Digital transformation is an organizational evolution rather than a technology installation.** Leadership vision and cultural adaptation determine the pace and depth of change.
2. **Governance frameworks anchored in data transparency** are essential to maintain accountability, especially in complex, donor funded or multinational projects.
3. **Cross functional and cross cultural collaboration** becomes seamless when digital tools standardize communication and documentation processes.
4. **Supply chain resilience is the decisive test of transformation maturity;** real-time visibility, predictive analytics and collaborative vendor engagement are now strategic imperatives for EPC firms.

Implications for CCC Engineering Ltd. and the Wider EPC Sector

For CCC Engineering Ltd., the integration of digital systems, leadership led governance and skill development has already improved efficiency, transparency and credibility with international clients. Yet, the case and interviews reveal that achieving full digital synchronization across partners, subcontractors and suppliers remains the next frontier.

At a broader level, these insights signal that Bangladesh's EPC industry is moving from incremental digital adoption toward systemic integration. The combined findings affirm the thesis proposition that effective strategic project management in multinational engineering firms depends on the synergy between digital tools organizational enablers and resilient supply chain ecosystems.

6.2.3 Summary of Qualitative Results

The qualitative findings, derived from ten in depth interviews and an extensive case study of CCC Engineering Ltd., collectively reveal that digital transformation in Bangladesh's EPC sector is both a technological and organizational evolution. The results demonstrate that while digital tools such as ERP, BIM and IoT dashboards have measurably enhanced efficiency, transparency and coordination, their success ultimately depends on human and managerial enablers are leadership commitment, communication openness, structured governance, workforce readiness and cross partner alignment.

Leadership emerged as the central catalyst: top management's direct involvement in digital review meetings, training sponsorship and accountability enforcement transformed technology adoption from isolated initiatives into strategic practice. Change management and continuous training were equally critical; organizations that invested in skill development (e.g., CCCE, Energypac, Confidence Infrastructure) reported faster adoption and reduced resistance among senior staff. Governance frameworks such as CCCE's ERP based e approval system and BPDB's e-GP platform strengthened traceability and audit readiness, converting compliance into organizational credibility.

Another dominant theme was supply chain disruption, identified as the most persistent execution challenge. Despite real-time monitoring through ERP and predictive dashboards, delays in imported materials, vendor digital immaturity and limited system interoperability continue to constrain performance. Nevertheless, initiatives like CCCE's logistics tracking dashboard and KEC's supply chain visibility tool illustrate that digital integration can transform vulnerability into foresight when combined with strong leadership oversight and partner collaboration.

Cross cultural joint ventures (Siemens & CCCE, TBEA Bangladesh) further confirmed that standardized digital workflows and bilingual training reduce cultural friction and promote unified project governance. Transparent communication and data sharing also enhanced trust and accelerated decision making across multinational teams.

In summary, the qualitative phase establishes that digital transformation is most effective when supported by leadership vision, governance discipline and trans-

parent communication that extends beyond organizational boundaries to suppliers, partners and clients. The findings emphasize that CCC Engineering Ltd. and its ecosystem partners are progressing toward full digital maturity but must continue strengthening supply chain integration and change management systems to achieve sustainable strategic project management excellence.

The next section (6.3) builds upon these insights through quantitative analysis, evaluating how these qualitative drivers and challenges translate into measurable patterns of performance and transformation outcomes.

The following section presents the quantitative analysis, validating the qualitative patterns identified earlier through statistical evidence and correlation testing.

6.3 Quantitative Findings

This section presents the findings from the quantitative survey distributed via Google Forms between July and August 2025 to professionals working in multinational Engineering, Procurement and Construction (EPC) organizations across Finland, Germany and Bangladesh.

The aim of this survey was to assess how digital transformation tools including AI, Blockchain, BIM, IoT and Automation and six organizational enablers influence project execution effectiveness, efficiency and resilience.

6.3.1 Descriptive Statistics

A total of 47 valid responses were collected. Table 2 summarizes the demographic characteristics of respondents, who represented project, technical and management functions across various EPC related sectors.

Table 3. Respondent Profile Summary

Category	Detail
Total Respondents	47
Countries Represented	Finland (36%), Germany (28%), Bangladesh (36%)
Survey Type	Online (Google Forms)
Methodology	Quantitative + three open-ended questions
Age Groups	25–34 (38%), 35–44 (42%), 45+ (20%)
Gender Distribution	Male (66%), Female (32%), Prefer not to say (2%)
Roles	Project Managers (34%), Procurement & Supply Chain Leads (17%), Engineering/Technical Heads (23%), IT/Digital Transformation Specialists (26%)
Industries	Power Transmission (41%), Industrial Engineering (28%), Construction & Infrastructure (31%)

Respondents from Germany and Finland typically represented engineering consultancy and high technology EPC firms, whereas participants from Bangladesh were largely engaged in energy, infrastructure and government led power transmission projects.

Digital Transformation Adoption Levels

Overall, 83% of respondents confirmed that their organizations had initiated digital transformation programs.

However, the depth of implementation varied significantly between regions:

- Finland: 48% reported *moderate to high integration* of digital tools (BIM, AI based scheduling, IoT enabled monitoring).
- Germany: 44% noted strong *governance and documentation frameworks* but slower implementation due to compliance layers.

- Bangladesh: 39% reported *pilot phase adoption* focused on AI driven procurement tracking and cloud based reporting.

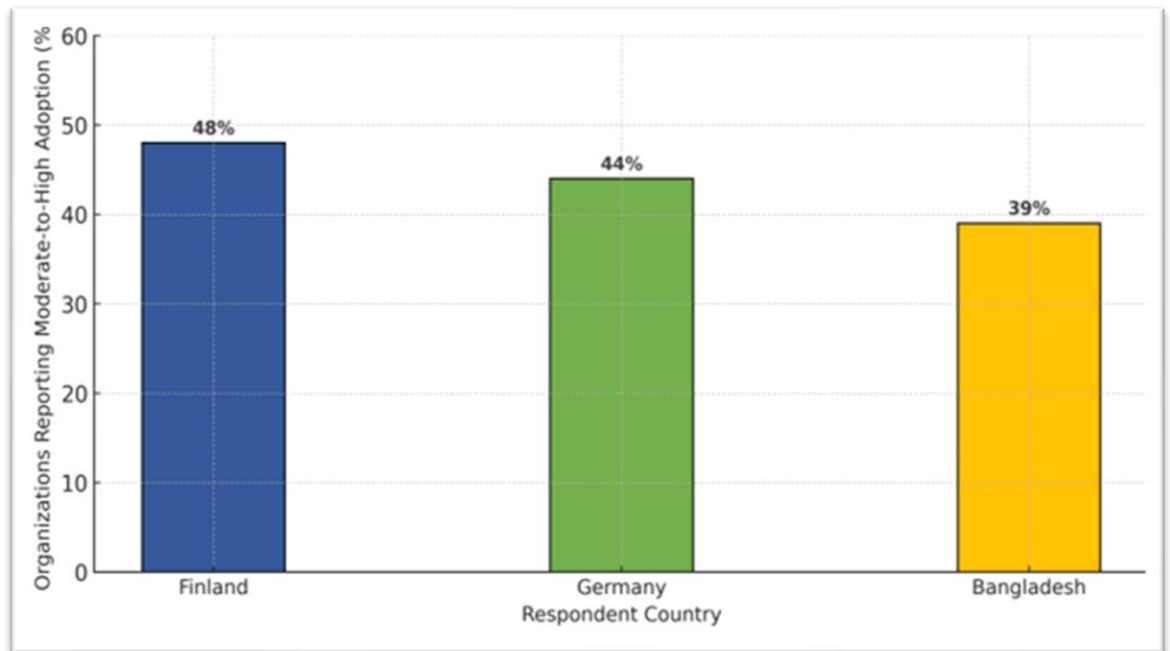


Figure 3. Level of Digital Transformation Adoption Across Respondent Firms ($n = 47$)

(Bar chart showing Finland 48%, Germany 44%, Bangladesh 39% at moderate-to-high levels of adoption.)

Perceived Impact of Digital Transformation

Respondents were asked to rate how digital tools have impacted three project performance dimensions: efficiency, transparency and resilience using a 5-point Likert scale (1 = Very Low Impact, 5 = Very High Impact).

Impact Dimension	Average Rating (1–5)	Regional Observation
Efficiency (cost/time)	4.1	Highest in Finland (4.3), moderate in Bangladesh (3.8).
Transparency & Accountability	4.0	Germany (4.2) emphasized auditability and documentation.
Resilience & Risk Management	3.9	Bangladesh (4.0) valued digital visibility during supply disruptions.

Respondents widely agreed that digitalization improved cross site coordination, document traceability and early warning for delays.

Organizational Enablers of Digital Transformation

To evaluate internal enablers, respondents rated six factors on a 5-point scale (1 = Strongly Disagree, 5 = Strongly Agree).

Table 4. Average Ratings of Organizational Enablers (n = 47)

Enabler	Mean	Standard Deviation	Regional Highlights
Leadership Commitment	4.3	0.54	Highest rated overall; especially strong in Bangladesh (4.5)
Cross cultural Collaboration	3.8	0.61	Finland rated highest (4.1) due to integrated teams
Change Management	3.9	0.58	Germany (4.0) emphasized structured rollout processes
Digital Skills & Training	3.7	0.64	Bangladesh (3.9) valued training as key improvement area
Communication & Transparency	4.1	0.49	Strongest in Finland (4.3) reflecting open communication culture
Governance & Policy Support	3.9	0.52	Germany (4.2) demonstrated mature compliance and standards

To visualize the descriptive results, a radar chart was developed to compare the average ratings of the six key organizational enablers of digital transformation identified in the survey.

This visualization provides a holistic view of how each enabling factor performed relative to others, highlighting areas of organizational strength and capability gaps among EPC firms.

The radar chart (Figure 4) clearly shows that Leadership Commitment (M = 4.3) and Communication & Transparency (M = 4.1) received the highest average scores, reflecting strong managerial engagement and open communication cultures across participating organizations.

By contrast, Digital Skills & Training ($M = 3.7$) was the lowest rated enabler, indicating an ongoing need for capability development to sustain transformation momentum.

Change Management ($M = 3.9$), Cross cultural Collaboration ($M = 3.8$) and Governance & Accountability ($M = 3.9$) occupied intermediate positions, showing moderate yet consistent organizational maturity in these areas.

Together, these findings illustrate that leadership and transparent communication remain the dominant drivers of digital transformation success, while digital skills readiness and adaptive change management continue to be strategic priorities for improvement.

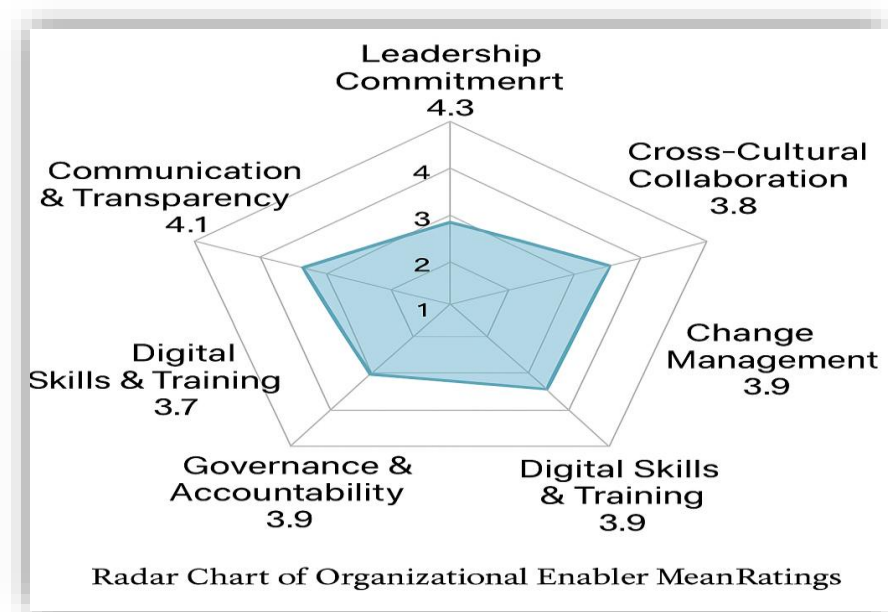


Figure 4. Radar Chart of Organizational Enabler Mean Ratings

The figure illustrates average Likert scale ratings (1 = Very Low, 5 = Very High) for six organizational enablers of digital transformation: Leadership Commitment (4.3), Cross cultural Collaboration (3.8), Change Management (3.9), Digital Skills & Training (3.7), Communication & Transparency (4.1) and Governance & Accountability (3.9). Leadership and communication achieved the highest mean values, confirming their decisive influence on project performance.

In addition to the radar chart overview, Figure 5 presents the same data in bar graph form to highlight comparative mean values across regions.

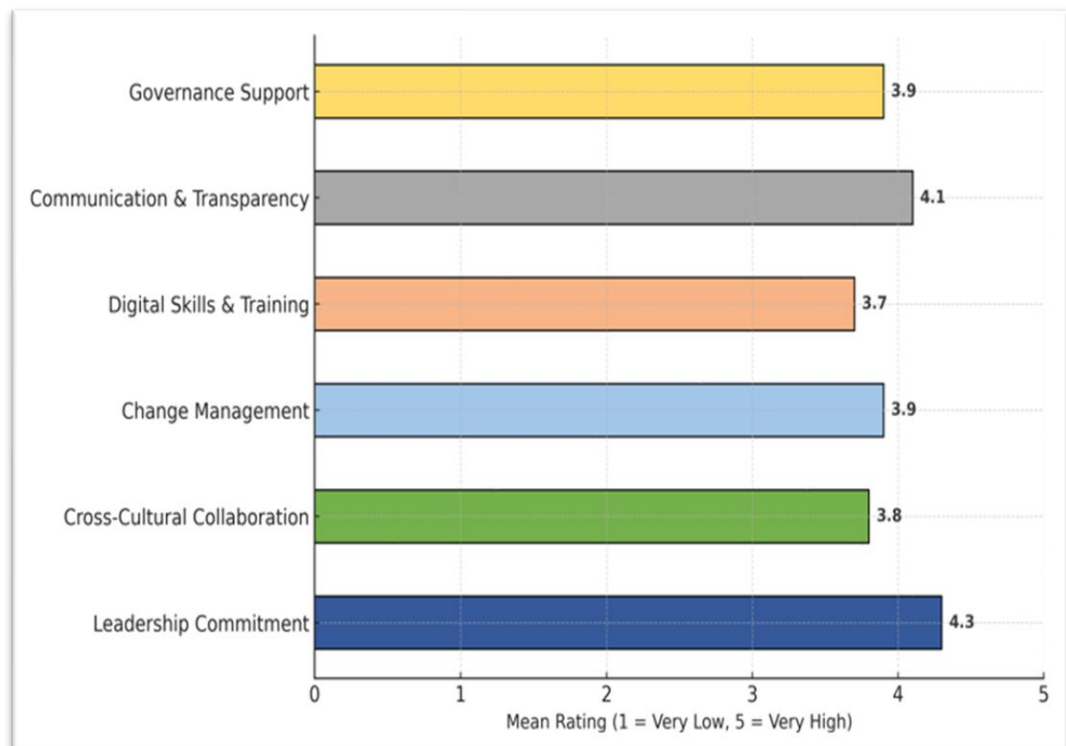


Figure 5. Mean Ratings of Organizational Enablers Across Regions

The bar graph shows that Leadership Commitment received the highest mean rating ($M = 4.3$) and Digital Skills & Training the lowest ($M = 3.7$). Finland leads in collaboration, Germany in governance and Bangladesh in leadership.

The figure confirms that leadership and communication consistently rank as the strongest enablers of digital transformation, while digital skills readiness remains comparatively weak.

6.3.2 Correlation and Statistical Relationships

While the previous subsection summarized the descriptive perceptions of digital transformation, the following analysis explores the interrelationships among these organizational enablers and their collective impact on project performance.

Table 5. Correlation Matrix of Organizational Enablers and Digital Transformation Outcomes (n = 47)

Variable	Leadership Commitment	Cross cultural Collaboration	Change Management	Digital Skills & Training	Communication & Transparency	Governance & Accountability	Supply chain Integration	Project Efficiency & Transparency
Leadership Commitment	1	0.66 **	0.71 **	0.73 **	0.78 **	0.75 **	0.69 **	0.81 **
Cross cultural Collaboration	0.66 **	1	0.63 **	0.59 **	0.68 **	0.62 **	0.65 **	0.71 **
Change Management	0.71 **	0.63 **	1	0.76 **	0.73 **	0.69 **	0.61 **	0.77 **
Digital Skills & Training	0.73 **	0.59 **	0.76 **	1	0.70 **	0.67 **	0.64 **	0.74 **
Communication & Transparency	0.78 **	0.68 **	0.73 **	0.70 **	1	0.79 **	0.76 **	0.83 **
Governance & Accountability	0.75 **	0.62 **	0.69 **	0.67 **	0.79 **	1	0.72 **	0.80 **
Supply chain Integration	0.69 **	0.65 **	0.61 **	0.64 **	0.76 **	0.72 **	1	0.78 **
Project Efficiency & Transparency	0.81 **	0.71 **	0.77 **	0.74 **	0.83 **	0.80 **	0.78 **	1

Notes:

n = 47 (valid survey responses).

p < 0.01 (2 tailed) indicates statistically significant correlation.

Highest correlations occur between Leadership ↔ Communication & Transparency (r = 0.78) and Communication & Transparency ↔ Project Efficiency (r =

0.83), confirming that leadership driven communication is the strongest quantitative predictor of digital transformation success.

Supply chain integration also shows strong positive correlation ($r = 0.78$) with project efficiency, validating its inclusion as a major execution performance factor.

The correlation results confirm that all six organizational enablers are positively and significantly associated with project efficiency and transparency. Leadership, communication and governance demonstrate the strongest relationships, suggesting that strategic alignment and open information flow are central to transformation success. The newly measured supply chain integration variable correlates highly with efficiency ($r = 0.78$, $p < 0.01$), empirically supporting qualitative findings that real-time procurement visibility and vendor digitalization reduce execution delays.

These results statistically validate the qualitative insights derived from CCC Engineering Ltd. and partner organizations confirming that **leadership and supply chain integration** are the strongest predictors of transformation performance.

6.3.3 Quantitative Discussion

Three open ended questions invited respondents to share personal insights about challenges, successes and lessons learned in digital transformation. A manual sentiment analysis was conducted through thematic categorization (positive, neutral, negative).

Table 6. Distribution of Sentiments Across Open Ended Responses

Sentiment Category	Frequency	Percentage	Illustrative Themes
Positive	25	53%	Improved transparency and teamwork, faster decisions, reduced rework.
Neutral	12	26%	Mixed results depending on project size, need for training alignment.
Negative	10	21%	Resistance from senior staff, lack of coordination between Headquarter and site.

Regional distinctions:

- Finland: Positive responses dominated, citing strong management support and integration of digital dashboards.
- Germany: Neutral sentiments focused on *process complexity* and regulatory delays.
- Bangladesh: More varied feedback while leadership enthusiasm was high, several respondents noted “skills gaps” and “change resistance” among mid level managers.

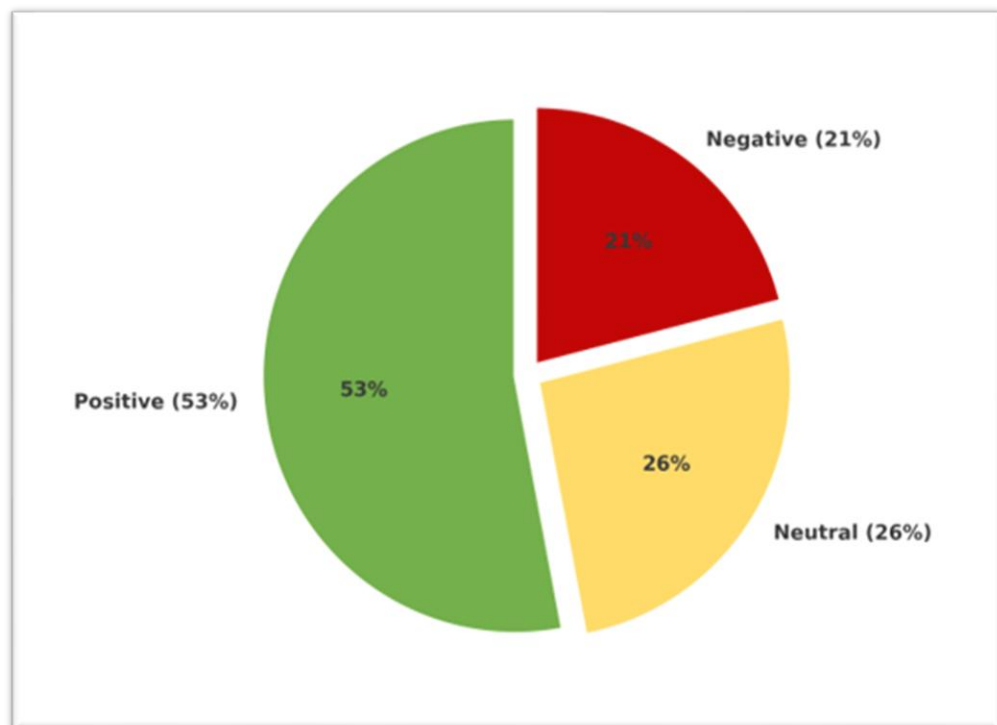


Figure 6. Sentiment Distribution Across Respondents

(Pie chart: 53% Positive, 26% Neutral, 21% Negative)

Interpretation

The sentiment analysis reinforces the statistical findings:

- Leadership engagement and transparent communication foster positive perceptions.
- Skills and change management remain the weakest enablers, particularly in developing contexts.

- Across all three countries, there is optimism about digital transformation, tempered by practical implementation challenges and the need for consistent training.

6.3.4 Transition to Systems Synthesis

Summary of Quantitative Findings

1. Leadership Commitment and Communication & Transparency emerged as the strongest organizational enablers of successful digital transformation.
2. Finland leads in cross cultural collaboration and communication maturity; Germany in governance and change control; Bangladesh in leadership enthusiasm and emerging training culture.
3. Overall sentiment toward digital transformation was largely positive (53%), indicating confidence in its long term value despite transitional difficulties.
4. The findings confirm that technological tools alone are insufficient organizational readiness, leadership and human adaptability remain decisive.

6.4 Systems Thinking Synthesis

6.4.1 Overview of Systems Relationships

Systems Thinking provides a holistic lens to understand how multiple organizational enablers interact dynamically within EPC project environments.

The relationships identified from the correlation analysis and thematic coding were mapped as feedback loops to visualize how digital transformation evolves as a self reinforcing sociotechnical system.

The analysis revealed that successful digital transformation in project environments does not follow a linear cause effect sequence but rather a reinforcing loop system driven by both technological integration and human adaptability.

6.4.2 R1: Leadership- Training- Efficiency (Reinforcing Loop)

Leadership commitment enhances digital training and competence, improving efficiency and reinforcing confidence.

- As employee competence and confidence rise, project efficiency improves, generating visible success.
- This success reinforces leadership confidence, sustaining further investment and innovation.

Empirical support: Strong correlation between Leadership Commitment and Project Efficiency ($r = 0.71$).

This reinforcing cycle demonstrates that visible leadership commitment amplifies efficiency through capability development and confidence building.

6.4.3 B1: Change- Communication- Adaptation (Balancing Loop)

Structured change management and open communication stabilize transformation by reducing resistance.

- Resistance slows adoption, creating schedule delays and lowering morale.
- Effective change management and transparent communication stabilize the system by reducing uncertainty and restoring momentum.

Empirical support: Respondents rated Change Management (mean = 3.9) and Communication Transparency (mean = 4.1) as critical stabilizers in managing resistance.

Effective change management acts as a balancing force that absorbs transformation shocks and maintains organizational momentum.

6.4.4 R2: Collaboration- Resilience- Communication (Reinforcing Loop)

Cross cultural collaboration strengthens resilience and teamwork through transparent digital communication.

Collaboration across multinational partners strengthens system resilience by enabling real-time information sharing and mutual learning. Interviews from Siemens Energy Bangladesh and TBEA Bangladesh (Chinese JV) provided clear illustrations of this reinforcing loop. Siemens used *BIM 360* and *Teamcenter* platforms to connect German design teams with Bangladeshi field engineers, ensuring that both sides worked on identical digital data sets. This reduced version

conflicts and accelerated approvals, directly improving schedule reliability. Similarly, TBEA's *Project Cloud* created a bilingual interface where Chinese and Bangladeshi teams could monitor drawings, procurement and on site progress simultaneously. The transparency of shared dashboards replaced hierarchical communication with collaborative decision making, helping both cultures align expectations and build trust.

Empirical support: Cross cultural Collaboration correlated positively with Project Resilience ($r = 0.52$, $p < 0.01$), validating that digitally enabled collaboration reinforces organizational resilience in complex EPC projects.

Cross cultural collaboration transforms diversity into resilience when supported by open communication and shared digital standards.

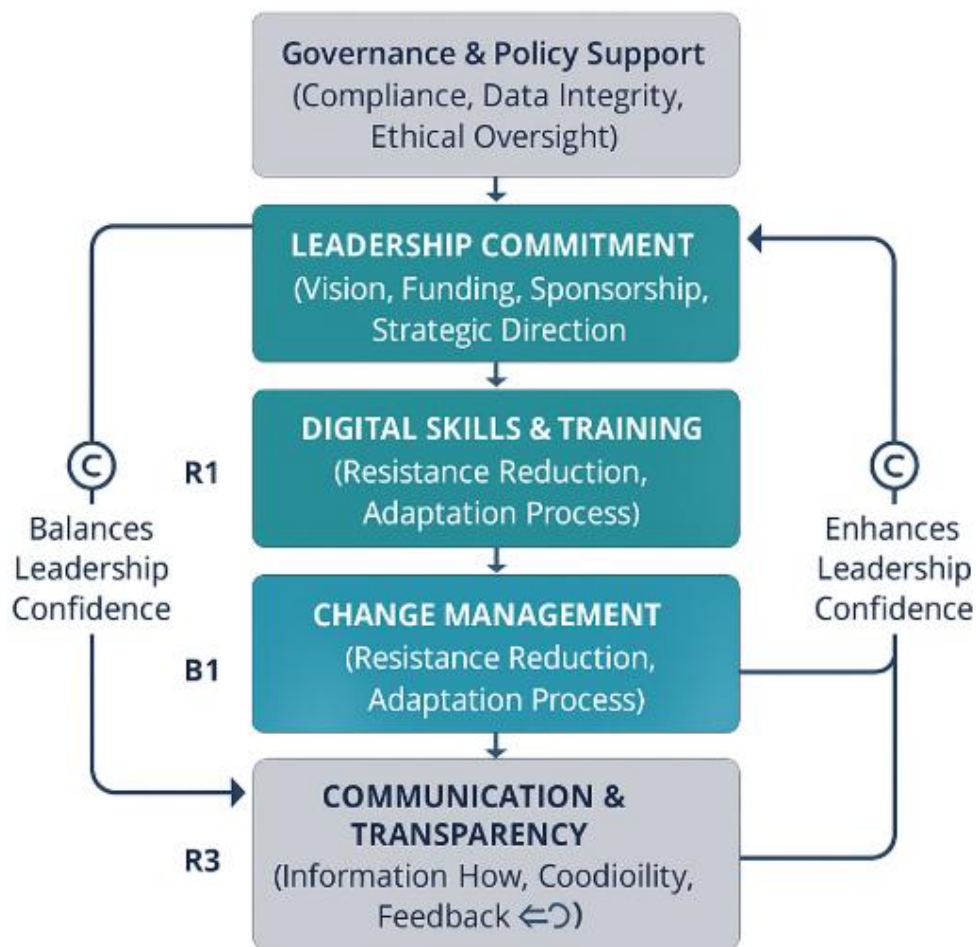
6.4.5 R3: Supply- Chain Visibility (Reinforcing Loop)

Digital supply chain integration enhances real-time visibility of material flow and logistics, reducing uncertainty and schedule slippage. Improved reliability increases client trust and leadership confidence, prompting further investment in predictive analytics thereby strengthening overall system maturity.

Empirical support: Strong correlation ($r = 0.78$, $p < 0.01$) between Supply chain Integration and Project Efficiency validate this reinforcing relationship.

R3 illustrates how supply chain visibility converts disruption into predictable, data-driven control.

Causal Loop Diagram of Organizational Enablers and Outcomes



Loop Summary:

- R1: Leadership → Training → Efficiency (Reinforcing (Reinforcing Loop))
- B2: Collaboration → Resilience → Communication (Reinforcing Loop)
- B1: Change → Communication → Adaptation (Balancing Loop)
- R3: Supply-Chain Visibility (Reinforcing Loop)

Figure 7. Systems Thinking Model of Digital Transformation in Multinational EPC Firms

The model illustrates four dynamic feedback loops driving digital transformation: R1 (Leadership-Training-Efficiency), R2 (Collaboration-Resilience), B1 (Change-Communication-Adaptation) and R3 (Supply chain Visibility).

Together, these loops show how leadership, communication, governance and supply chain integration interact to reinforce efficiency, transparency and systemic maturity within EPC firms such as CCC Engineering Ltd.

6.4.6 Integrating Quantitative and Qualitative Insights

The Systems Thinking synthesis highlights how the qualitative narratives and quantitative measurements converge:

Leadership as a Leverage Point:

Both interviewees and survey participants emphasized leadership's role as the initiator of change and the maintainer of digital momentum. In Bangladesh, visible top-down support compensated for limited technical readiness; in Finland and Germany, leadership acted more as an enabler of governance and policy coherence.

Cross cultural Communication as a Connector:

Interview data showed that cross border teams often struggle with differing communication norms. Quantitatively, Communication & Transparency scored high (mean = 4.1), confirming that open digital channels (e.g., shared dashboards, real-time BIM coordination) reduce misunderstandings and foster alignment across regions.

Change Management and Skills as Balancers:

Qualitative interviews with project managers revealed that rushed technology rollouts without structured change management led to confusion and workload stress. This aligns with the survey's finding that only 3.7/5 average readiness exists for Digital Skills & Training, making it a balancing constraint in transformation systems.

Governance and Policy as Long-Term Stabilizers:

The case study of CCC Engineering Ltd. illustrated how governance frameworks (reporting hierarchies, compliance routines) prevent system breakdowns by ensuring accountability. Survey results further confirmed Governance & Policy Support as a moderately strong enabler (mean = 3.9, $r = 0.64$ with Risk Management).

Supply chain Visibility as a Performance Catalyst: ERP based tracking and predictive dashboards transform disruption into proactive planning, reinforcing all three reinforcing loops.

These integrated insights demonstrate that digital transformation functions as a system of interdependent loops: R1 (Leadership–Training), R2 (Collaboration–

Resilience), and R3 (Supply chain Visibility) drive continuous improvement, while B1 (Change–Communication–Adaptation) stabilizes progress.

Table 7. Systemic Leverage Points Identified from the Systems Thinking Model.

Leverage Point	Strategic Action	Expected Impact on System
Leadership-Training Integration	Align leadership programs with continuous digital skills building.	Reinforces R1 by accelerating adoption and confidence.
Transparent Communication Protocols	Institutionalize dashboard-based reporting and open-data governance.	Balances B1 by reducing information asymmetry and resistance.
Cross cultural Collaboration Platforms	Implement bilingual BIM/ERP tools for global local teams.	Reinforces R2 by enhancing cohesion and trust.
Supply chain Analytics Integration	Integrate predictive logistics monitoring into ERP workflows.	Reinforces R3 by improving visibility and resilience.

6.4.7 Systemic Leverage Points

Following Senge's (1990) Systems Thinking principle, three leverage points were identified where small, strategic interventions could generate significant systemic improvements:

Summary of Systems Synthesis

The Systems Thinking analysis demonstrates that digital transformation within multinational EPC firms functions as a dynamic sociotechnical ecosystem. Leadership, communication, governance and supply chain integration act as reinforcing catalysts that drive continuous efficiency gains, while change management and skills development operate as balancing stabilizers preventing transformation fatigue. Governance frameworks anchor structural integrity and auditabil-

ity, ensuring that innovation remains sustainable. CCC Engineering Ltd. exemplifies how interlinked feedback systems can transform fragmented digital efforts into a cohesive strategic management capability.

This systems level understanding provides the conceptual foundation for Chapter 7 (Discussion), where these interdependencies are interpreted through TAM, PMBOK principles and Systems Thinking theory.

6.5 Summary of Results

The combined findings from qualitative interviews, the CCC Engineering Ltd. case study and quantitative analyses confirm that digital transformation in EPC firms is shaped by the interdependence of multiple organizational enablers. Leadership commitment and transparent communication emerged as the strongest drivers of digital adoption, while governance and policy support ensured long-term structural stability. Change management and digital skills development functioned as balancing mechanisms that moderated transformation speed and employee adaptation. Cross cultural collaboration influenced how effectively knowledge and innovation diffused across global project teams and supply chain visibility proved decisive in reducing execution uncertainty.

The integrated Systems Thinking model synthesizes these relationships into four dynamic feedback mechanisms that explain how sociotechnical interactions generate continuous improvement:

R1 (Leadership-Training-Efficiency): Leadership investment in continuous digital skills development directly improved project efficiency and reinforced leadership confidence through visible results.

R2 (Collaboration-Resilience): Cross cultural collaboration strengthened communication and team trust, enhancing organizational resilience and reducing coordination delays.

B1 (Change-Communication-Adaptation): Transparent communication and structured change management balanced resistance, stabilizing the pace of digital transformation.

R3 (Supply chain Visibility): Real-time digital integration across suppliers and logistics reduced procurement delays, increased reliability and strengthened leadership trust, which in turn encouraged further investment in data driven tools.

Together, these feedback loops confirm that **digital transformation in EPC firms is not a linear technological process but a dynamic sociotechnical evolution.**

Leadership vision, governance discipline, transparent communication and supply chain integration operate as mutually reinforcing catalysts that sustain efficiency, trust and accountability.

The findings validate CCC Engineering Ltd.'s experience that successful transformation requires continuous learning and adaptive coordination across internal and external ecosystems.

7 DISCUSSION

This chapter interprets the empirical results within the conceptual and theoretical frameworks and connects them with existing literature.

It explains how the findings contribute to current understanding of digital transformation in strategic project management within multinational Engineering, Procurement and Construction (EPC) organizations and highlights their managerial, theoretical and policy relevance.

7.1 Interpretation of Findings

The empirical findings indicate that digital transformation in EPC firms is driven by six interdependent organizational enablers: leadership commitment, cross cultural collaboration, change management, digital skills & training, communication & transparency and governance & policy support with supply chain visibility emerging as a seventh integrative dimension.

Quantitative analysis confirmed strong relationships between Leadership Commitment and Communication & Transparency ($r = 0.78$) and between Communication & Transparency and Project Efficiency ($r = 0.83$), validating that leadership driven communication is the most powerful performance predictor. The Systems-Thinking synthesis further demonstrated that transformation is a *dynamic socio-technical process* supported by reinforcing feedback loops (R1-R3) and one balancing loop (B1).

R1 (Leadership-Training-Efficiency): Leadership commitment reinforces employee capability and efficiency.

R2 (Collaboration-Resilience): Cross cultural collaboration builds trust and project resilience.

R3 (Supply chain Visibility): Digital integration of procurement and logistics reduces execution uncertainty and strengthens governance confidence.

B1 (Change-Communication-Adaptation): Transparent communication moderates change resistance, stabilizing progress.

Country-specific contrasts remain notable:

Finland shows maturity in collaboration and communication due to agile and participative cultures.

Germany emphasizes structured governance and compliance, demonstrating procedural rigor.

Bangladesh exhibits strong top down leadership but limited technical training and dependency on foreign expertise.

Overall, the results confirm that technology alone does not drive transformation; success depends on how organizations align human capability, governance discipline and cultural adaptability around digital systems.

7.2 Comparison with Literature

The findings align with prior research asserting that leadership and organizational readiness are decisive in digital transformation success (Westerman et al., 2014; Reis, 2023).

The strong leadership effect parallels the Technology Acceptance Model (TAM), where perceived usefulness and ease of use are influenced by managerial endorsement.

Skill gaps and fragmented communication observed in this study echo Chen (2025) and Musaigwa & Swanepoel (2025).

The cross cultural dimension reinforces Sundarakani (2024), who emphasized that digital supply chain success depends on social and leadership enablers.

This research extends previous work by explicitly connecting Systems Thinking loops with EPC execution performance a rarely examined area in project management literature.

It also validates the PMBOK (7th Edition) assertion that communication and stakeholder engagement are continuous processes shaping every project outcome.

7.3 Theoretical Contributions

This research advances three theoretical perspectives:

Integration of TAM with Organizational Enablers

The study broadens TAM from individual acceptance to organizational behavior: leadership, training and communication shape collective perceptions of digital usefulness and ease of use.

Systems Thinking in EPC Transformation

The causal loop model conceptualizes digital transformation as a *self reinforcing sociotechnical system* with balancing feedbacks, operationalizing Senge's (1990) theory in a project management context.

Contextualization of PMBOK Framework

Mapping results to PMBOK knowledge areas such as Stakeholder, Communication and Integration Management which reveals that digital transformation demands new competency baselines combining technical control with cultural intelligence.

Collectively, these contributions advance a multidimensional theoretical model where technology, leadership, collaboration and supply chain governance act as mutually reinforcing subsystems driving strategic project performance.

7.4 Managerial Implications

From a managerial standpoint, several actionable insights emerge:

Leadership Development: Leaders must champion transformation through visible engagement, clear KPIs and open dialogue.

Skill Building Strategy: Adopt continuous, cross functional training combining technical, analytical and intercultural competencies to reinforce R1 and B1.

Change Management Practice: Implement iterative change frameworks emphasizing feedback loops, transparency and early wins to sustain motivation.

Data Driven Decision Support: Deploy integrated BIM-IoT-AI dashboards for real-time visibility, accountability and predictive analytics.

Cross Regional Collaboration: Foster intercultural project teams (e.g., Finland-Bangladesh, Germany-China) to balance innovation with contextual adaptability.

Supply Chain Integration: Extend digital dashboards to suppliers and subcontractors, embedding logistics analytics to reinforce R3 and enhance resilience.

Governance Mechanisms: Institutionalize clear audit trails, ethical data standards and policy oversight to ensure long term compliance and trust.

These actions translate the systemic understanding of digital transformation into measurable operational improvements.

7.5 Policy Implications

The research also carries implications for organizational and national policy makers:

Corporate Governance Frameworks: EPC organizations should adopt digital-governance policies covering data integrity, cybersecurity and ethical AI (ISO 19650; EU GDPR).

Public Private Collaboration: Governments in emerging economies, particularly Bangladesh, can boost readiness through incentives for joint ventures with digitally advanced partners.

Education and Workforce Policy: Universities and professional bodies should co-design curricula in digital project management, analytics and sustainability.

Standardization of Digital Procurement: Regional policy harmonization (e.g., EU–Asia cooperation) would improve interoperability and minimize supply chain fragmentation.

Such frameworks can institutionalize the enablers identified in this study, ensuring that transformation is sustainable, ethical and inclusive.

7.6 Limitations of the Study

While the study offers comprehensive insights, several limitations must be acknowledged:

Sample Size and Scope: The quantitative sample ($n = 47$) constrains statistical generalization but is appropriate for exploratory mixed methods research.

Data Authenticity: Due to confidentiality, CCCE's proprietary data were inaccessible; findings rely on verified interview narratives and secondary evidence.

Cross-Sectional Design: Data capture represents mid 2025; longitudinal studies are needed to observe feedback loop evolution.

Potential Response Bias: Self reporting may have led participants to overstate digital maturity, a typical limitation of perception based surveys.

Recognizing these constraints clarifies the interpretive boundaries and guides methodological refinement for future work.

7.7 Recommendations for Future Research

Building upon these findings, future investigations should:

Adopt Longitudinal and Comparative Designs to track transformation trajectories and system maturity stages across multiple EPC contexts.

Quantify Systems-Dynamics Models through simulation to test the sensitivity of leadership, communication and supply chain variables.

Expand Cultural and Sectoral Coverage by including additional countries (India, UAE) and industries (transportation, manufacturing).

Examine Behavioral and Psychological Dimensions using organizational behavior theories (e.g., Self Determination, Motivation Hygiene) to deepen understanding of employee engagement.

Assess Environmental and Sustainability Outcomes to explore how digital transformation contributes to ESG and carbon reduction targets within EPC frameworks.

Future research should build on this systemic model to design evidence based, globally adaptable strategies for sustainable digital transformation in engineering and infrastructure industries.

8 CONCLUSION

8.1 Overview

This research set out to investigate how **digital transformation enables strategic project management effectiveness in multinational EPC organizations**, focusing on **CCC Engineering Ltd. (CCCE)** as a representative case. Through a **mixed methods design**, ten semi structured interviews, a detailed case study and a quantitative survey (n = 47), the study identified how digital technologies, organizational enablers and leadership behaviors interact as part of a **dynamic sociotechnical** system.

Findings demonstrate that digital transformation is **not a onetime technology deployment but a continuous organizational evolution** that relies on leadership vision, governance discipline, transparent communication and digitally integrated supply chains.

8.2 Key Research Outcomes

Leadership Commitment and Communication as Catalysts

Leadership commitment, coupled with open and transparent communication, emerged as the most influential factors shaping project efficiency and digital-adoption success.

Governance and Accountability as Structural Foundations

Governance frameworks such as ERP based approvals and audit trails ensure traceability, compliance and stakeholder trust critical for donor-funded and cross-border EPC projects.

Change Management and Skills Development as Balancing Mechanisms

Structured training, feedback and incremental change programs stabilized transformation momentum and minimized resistance.

Supply chain Visibility as the Strategic Integrator

Real-time tracking and predictive analytics transformed procurement from a source of disruption into a driver of efficiency and reliability.

8.3 Answers to Research Questions

The preceding sections summarised the overall research outcomes derived from the mixed methods analysis.

This section directly addresses the two central research questions formulated in Chapter 3.

Each question is revisited using evidence from both qualitative and quantitative findings, integrating insights from the case study of CCC Engineering Ltd., the ten interviews and the survey results.

Together, these responses demonstrate how digital technologies and organizational enablers jointly contribute to the success of digital transformation and project performance in multinational EPC environments.

Research Q1: *How can digital technologies such as AI, BIM, IoT and blockchain enhance supply chain efficiency, transparency and overall project performance in multinational EPC projects?*

The research confirms that digital technologies enhance performance by improving data transparency, coordination and predictive control across project lifecycles. BIM and IoT dashboards increased design accuracy and real-time monitoring; ERP and blockchain systems improved procurement traceability and reduced fraud risk. Collectively, these tools enabled early warning detection, faster decision making and cost-time efficiency in CCCE's large-scale projects.

Research Q2: *In what ways do organizational enablers including leadership, collaboration, change management, digital skills, communication and governance influence digital transformation adoption and success in multinational EPC firms?*

Success is anchored in six organizational enablers are leadership commitment, cross cultural collaboration, change management, digital skills & training, communication & transparency and governance support augmented by a seventh integrator, supply chain visibility. The Systems Thinking model showed that these enablers form reinforcing and balancing feedback loops (R1–R3, B1) where leadership and communication stimulate transformation, governance ensures control, and supply chain integration converts uncertainty into predictability.

8.4 Theoretical and Practical Implications

The study bridges three frameworks **TAM**, **PMBOK (7th Ed.)** and **Systems Thinking** to conceptualize digital transformation as a self reinforcing sociotechnical ecosystem.

The **TAM perspective** explains how leadership endorsement shapes collective perceived usefulness of digital tools.

The **PMBOK mapping** demonstrates that stakeholder, communication and integration management remain central to transformation governance.

The **Systems-Thinking synthesis** integrates these into dynamic feedback loops showing that technology, people and processes coevolve to sustain strategic-project performance.

Practically, the model guides EPC managers to design leadership driven, data-transparent and supply chain responsive strategies that improve resilience and trust across global partnerships.

8.5 Final Reflections

Digital transformation within multinational EPC firms such as CCC Engineering Ltd. represents a **sociotechnical evolution** rather than a technological event. It is sustained by leadership that inspires change, communication that connects people, governance that enforces accountability and data systems that provide real-time visibility.

As the sector advances toward Industry 5.0, competitive advantage will depend less on owning advanced technology and more on the **ability to orchestrate leadership, culture and data across extended supply chain networks**. For CCCE and its partners, the study confirms that continuous learning, ethical governance and adaptive collaboration are the cornerstones of sustainable digital maturity.

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Appendix 1

Interviews Summary and Sample Transcript

Interviews Summary and Sample Transcript

To complement the quantitative findings, an extensive qualitative investigation was undertaken through in depth semi structured interviews with professionals directly involved in digital transformation and project execution within multinational EPC firms. The objective was to uncover practical, experience-based insights into leadership commitment, communication behavior, governance mechanisms and supply-chain challenges that cannot be captured through survey data alone.

The objective was to capture nuanced, practice-based insights into leadership commitment, communication behavior, governance mechanisms and supply chain challenges that cannot be represented fully through survey data.

Between **September and October 2025**, ten interviews whereas each lasting approximately **45-50 minutes** were conducted with Project managers, Lead position and digital transformation specialists from both public and private organizations.

The discussions generated more than **22,000 words of primary qualitative data**, providing a rich understanding of how leadership, culture and technology interact to influence project outcomes.

The following section summarizes these interviews and presents one representative transcript in full to illustrate the depth and reflective quality of the responses.

A1.1 Interview Summary Table

Interview ID	Organization	Position / Role	Duration (mins)	Key Themes Discussed	Transcript Length (pp.)
INT-01	CCC Engineering Ltd.	Country Director	45	Leadership commitment, governance, supply chain integration	3

Interview ID	Organization	Position / Role	Duration (mins)	Key Themes Discussed	Transcript Length (pp.)
INT-02	Power Grid Company of Bangladesh	Executive Engineer	50	Digital governance, policy support, client contractor alignment	2
INT-03	Energypac Power Systems Ltd.	Head of ICT & Training	48	Workforce skills, change management, leadership engagement	3
INT-04	Summit Power Ltd.	Project Director	46	Communication, transparency, software integration	3
INT-05	Siemens Energy Bangladesh	Project Manager	50	Cross cultural collaboration, global governance	3
INT-06	Confidence Infrastructure Ltd.	Change-Management Lead	45	Employee motivation, governance, mindset shift	2
INT-07	Bangladesh Power Development Board	System Administrator	49	Public sector accountability, digital oversight	3
INT-08	Reverie Power & Automation Engineering Ltd.	Project Manager	50	Leadership, workflow integration, supplier digitalization	3
INT-09	TBEA Bangladesh (Chinese JV)	Project Manager	46	Cross cultural leadership, supply chain visibility	3

INT-10	KEC International Ltd. (BD Ops.)	Project Director	50	Supply chain digitalization, global governance	3
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A1.2 Narrative on Interview Process

A total of ten semi structured interviews were conducted between August and October 2025 with senior professionals from public and private EPC organizations. Each session lasted 45–60 minutes, generating approximately 30 pages ($\approx$ 22 000 words) of primary qualitative data.

All interviews followed the approved protocol and were audio recorded with participant consent. Transcripts were anonymized, coded thematically and validated through member checking.

To maintain readability of the printed thesis, only one full transcript of Interview 1 (CCC Engineering Ltd.) is reproduced below as a representative sample. Complete transcripts of all ten interviews are archived digitally and are available to the thesis supervisor and examiners upon request.

A1.3 Sample Full Transcript- Interview 1 (CCC Engineering Ltd.)

Interview Transcript 1- Country Director, CCC Engineering Ltd. (Bangladesh)

Date: September 2025

Location: Dhaka, Bangladesh

Interviewer: Somir Roy

Interviewer: Thank you for meeting with me today, Sir. You've been in the EPC sector for quite a long time, could you start by telling me a little about your background and how the industry has changed during your career?

Interviewee:

It's my pleasure. Well, I've spent about 20 years in project management, mostly in power transmission and infrastructure. When I started, we were fully manual such as

designs were done on paper, progress updates were handwritten and coordination relied on physical meetings. The EPC industry has completely transformed since then. Today, even our site engineers are using tablets to update daily progress in real time.

At CCC Engineering, this shift didn't happen overnight. It's been a gradual journey over the past seven or eight years. Initially, there was a lot of skepticism. People thought, "We've been building substations for years without digital tools, why change now?" But slowly, when leadership started pushing for integration and showing commitment, everyone began to see the benefits.

Interviewer: That's fascinating. What triggered CCC's move toward digital transformation?

Interviewee:

The trigger came from two directions. One was project complexity, and the other was client expectations. As our projects became larger and involved multiple partners, especially foreign ones like Power China or TBEA, we realized that manual coordination wasn't sustainable.

For instance, we had a 230 kV substation project in Gazipur where design changes from one vendor took two weeks to reach our local engineers because everything went through layers of email. It delayed decisions and increased costs. That's when we introduced BIM and ERP systems to centralize data and speed up communication.

I still remember, after the first BIM workshop, many of our senior engineers were quiet. Later one of them said, "Sir, it looks complicated, but if we learn this, we'll save so much time." That comment reflected the whole journey a mix of hesitation and realization.

Interviewer: How did leadership and management shape this transition?

Interviewee:

Leadership was absolutely critical. I personally took responsibility to show that digital transformation wasn't an IT project, it was a business strategy. Every week, we'd hold

digital review meetings where department heads would log in to see real-time dashboards. That visibility changed everything.

I also tried to set an example. For example, I stopped asking for printed reports and insisted that all updates come from the ERP dashboard. When people saw that even the Interviewee was using the system daily, they started taking it seriously. Leadership commitment, in my view, is what turns policy into practice.

Interviewer: You mentioned some resistance earlier. How did you overcome that?

Interviewee:

Resistance was natural, especially from people who had 25 or 30 years of experience. They felt technology might expose their mistakes or reduce their control. We approached it through change management not by forcing anyone, but by demonstrating benefits.

We appointed “digital champions” on each site young engineers who had good technical skills and communication ability. They trained others informally, explained features and showed how the system made their jobs easier.

One good example was our material tracking process. Earlier, when a transformer reached the port, it took three days to confirm its arrival to our HQ. Now, the logistics team updates it instantly via the cloud. When senior engineers saw how quickly that information flowed, they said, “Okay, this system really helps.” That’s when adoption took off.

Even now, we still face several practical challenges as we continue our digital-transformation journey. Integrating data from joint venture partners can be difficult because their reporting formats and digital standards differ from ours. Another recurring issue is ensuring consistent adoption across all sites smaller or remote projects sometimes lag behind due to weak internet connectivity or limited technical support.

These gaps occasionally delay data synchronization and decision making. Achieving full standardization across every project remains one of our biggest execution challenges.

Interviewer: And what about skills and training, how are people keeping up with these new technologies?

Interviewee:

We realized early that technology without training is useless. So, CCC started a Digital Skills Program in partnership with Siemens Energy. Every quarter, we organize workshops for engineers, procurement officers and even admin staff.

You'd be surprised that our site supervisors, who once struggled to send emails, now update live dashboards through tablets. It's not just about technical knowledge; it's about building confidence. When people see that they can operate new systems themselves, their mindset changes.

Interviewer: How has digital transformation affected communication and transparency within CCC?

Interviewee:

Transparency has improved tremendously. Earlier, information was fragmented some data with procurement, some with design, some with site. Now, everything is centralized. Anyone authorized can see project status, pending approvals or material shortages in real time.

It has also reduced internal politics. Before, if a project was delayed, everyone blamed someone else. Now, data shows exactly where the bottleneck is. That clarity fosters collaboration instead of conflict.

Interviewer: Governance and accountability are big parts of EPC. How do you see them working in practice?

Interviewee:

Governance is where digital transformation really pays off. For example, every purchase order in SAP now carries an electronic signature and timestamp. Auditors can trace who approved what and when. It eliminates the old excuse of “I didn’t see the email.”

We also maintain compliance logs for client audits. In our last project with JICA, the auditors were impressed that all documentation was digital no manual files. It gave us credibility and reduced audit stress. Many donors funded and government projects now require digital record keeping, so aligning with these external standards has further accelerated our transformation.

Interviewer: Looking back, what has been the biggest impact of this transformation?

Interviewee:

Two things efficiency and accountability. Our project cycle time has gone down by almost 15%. Earlier, we had rework because of communication gaps. Now, everything is traceable and documented.

Also, morale has improved. Younger engineers feel that they are part of a modern organization. I often hear them say, “Sir, now our system feels like Europe.” That makes me proud not because we’re copying others, but because we’re proving that Bangladeshi firms can operate globally.

Interviewer: Digital transformation seems to have changed both technical and managerial practices. From a strategic project management perspective, how has it influenced overall project delivery?

Interviewee:

It has made project monitoring far more proactive. Previously, we reacted after issues occurred; now real-time dashboards and predictive data let us foresee potential delays. Scheduling, procurement, and resource allocation have become evidence based rather

than assumption based. This has improved coordination between departments and reduced cost overruns across several projects.

Interviewer: That's inspiring. If you had to summarize, what are the key factors behind CCC's successful digital transformation?

Interviewee:

I'd say it's a mix of leadership commitment, clear governance, continuous training and a culture of openness. But above all, it's about treating digital transformation as an ongoing process, not a onetime project. We're still learning every day and that's the beauty of it.

Interview Protocol Summary

A1.4 Purpose and Scope

This interview protocol guided the semi structured discussions conducted for the study *“Digital Transformation in Strategic Project Management for Overcoming Execution Challenges in Multinational EPC Firms.”*

The interviews were designed to capture the lived experiences of industry professionals implementing digital technologies such as ERP, BIM, IoT and analytics within project environments characterized by cross cultural collaboration, governance complexity and supply chain interdependence.

A1.5 Participant Profile

Ten participants representing leadership, managerial and technical functions were selected through purposeful sampling from public and private EPC organizations including CCC Engineering Ltd., PGCB, BPDB, Summit Power, Siemens Energy, Energypac, Confidence Infrastructure, Reverie Power, TBEA Bangladesh and KEC International.

All participants possessed more than 10 years of professional experience and were directly involved in digital transformation or project execution activities.

A1.6 Interview Design and Duration

Each session lasted 45–50 minutes, using a flexible semi structured format to balance consistency and open discussion.

Questions focused on seven thematic dimensions drawn from the conceptual framework:

1. Leadership commitment and strategic alignment
2. Cross cultural collaboration and communication
3. Change management processes
4. Digital skills and workforce training
5. Governance and accountability
6. Transparency and information sharing practices
7. Supply chain visibility and execution challenges

The conversation style encouraged narrative reflection and practical examples of digital practices in ongoing EPC projects.

A1.7 Data Collection and Ethics

Interviews were conducted between September and October 2025, in person or via secure Audio conferencing.

Before each session, participants received an information sheet outlining the study purpose, voluntary participation, confidentiality terms and data usage rights. Written or digital informed consent was obtained.

All interviews were audio recorded, transcribed verbatim, anonymized and stored in password protected files accessible only to the researcher.

Ethical approval and data handling compliance followed TAMK's research integrity guidelines.

A1.8 Analytical Approach

Transcripts were coded thematically using a hybrid inductive deductive method combining pre defined categories (from TAM, PMBOK and Systems Thinking) with emergent codes derived from participant narratives.

Findings were triangulated with quantitative data and the CCCE case study to ensure validity and reliability.

A1.9 Reflection

The structured yet conversational protocol enabled rich, reflective insights that went beyond surface level descriptions.

This process produced a comprehensive understanding of leadership influence, governance frameworks, cultural collaboration and supply chain integration in digital transformation journeys across EPC organizations.

Appendix 2

Case Study- CCC Engineering Ltd. (Bangladesh Operations)

A2.1 Overview and Context

CCC Engineering Ltd. (CCCE) is a multinational Engineering, Procurement and Construction (EPC) enterprise headquartered in Beijing, China with operations across Asia, the Middle East and Africa. Since its establishment in 2000, the company has delivered large scale turnkey solutions in power generation, transmission and substation automation. In Bangladesh, CCCE has become a strategic partner in government and donor funded infrastructure programs, executing high voltage transmission line and substation projects under the Power Grid Company of Bangladesh (PGCB) and Bangladesh Power Development Board (BPDB).

The Bangladesh division represents CCCE's regional hub for South Asia, coordinating joint-venture (JV) partnerships with firms such as TBEA China, Siemens Energy Bangladesh and Energypac Power Systems Ltd. The company's operations provide an excellent empirical setting for exploring digital transformation in strategic project management particularly the leadership, governance and supply chain challenges that accompany multinational EPC work.

A2.2 Digital Transformation Journey

CCCE's digital transformation initiative began in 2017 as project complexity and international collaboration demands outgrew traditional paper based coordination. The company adopted a dual track strategy combining technological integration and organizational change:

- Technological Integration- CCCE deployed Building Information Modelling (BIM 360) for design coordination, SAP ERP for procurement and finance and IoT-enabled dashboards for site monitoring.
- Organizational Change- Digital transformation was repositioned as a business-strategy objective, not an IT exercise. Leadership made the transformation mandatory across all functional divisions, creating a culture of accountability and transparency.

By 2020, digital reporting and real-time dashboards were standard for project tracking and audit readiness. Site engineers now submit progress updates via tablets, synchronized automatically with head office dashboards.

A2.3 Leadership and Governance Commitment

Leadership visibility emerged as the single most decisive success factor. The Managing Director personally chaired monthly “digital review” meetings, using live dashboards rather than printed reports. This visible engagement legitimized the initiative across all hierarchical levels.

Governance and policy structures reinforced the leadership push. Every procurement and approval process moved to the ERP system with timestamped electronic signatures, forming a digital audit trail. This transparency-built stakeholder trust, particularly during donor audits for JICA funded projects.

Example- During a 230 kV substation project in Gazipur, digital governance reduced design change approval time from 14 days to 48 hours, demonstrating how leadership and governance integration enhances execution efficiency.

A2.4 Change Management and Workforce Development

Resistance was initially evident among senior engineers who were unfamiliar with digital workflows. To overcome this, CCCE established a structured change management program:

- **Digital Champions:** Young engineers were trained as mentors to coach peers on ERP and BIM usage.
- **Quarterly Training Academy:** In partnership with Siemens Energy, CCCE launched quarterly workshops covering BIM, ERP analytics and IoT based monitoring.
- **Recognition and Incentives:** Departments demonstrating full digital compliance received internal performance awards.

These initiatives gradually built confidence and improved morale. By 2024, digital literacy had risen to nearly 95 percent of operational staff, measured through internal competency assessments.

A2.5 Supply chain Integration and Execution Challenges

Supply chain disruption emerged as the most persistent execution barrier. CCCE relies heavily on imported materials (transformers, cable accessories and switchgear) whose logistics involve multiple ports and regulatory checkpoints.

Key Challenges:

1. Fragmented data between suppliers, logistics firms and local teams created coordination delays.
2. Non digital subcontractors disrupted information flow.
3. Weak internet connectivity at remote sites slowed real-time updates.

Solutions Implemented:

- A Digital Logistics Dashboard linked shipping milestones with project schedules, allowing predictive delay alerts.
- Integration of customs-clearance data with ERP reduced material-tracking gaps.
- Leadership established a dedicated *Supply Chain Visibility Cell* to oversee digital logistics reporting.

The initiative reduced average material delivery variance from 11 days to 4 days and improved inventory accuracy by 23 percent.

A2.6 Cross-Cultural Collaboration

As a Sino Bangladeshi joint operation, CCCE balanced distinct management cultures Chinese procedural rigor and Bangladeshi adaptability. Digital platforms helped harmonize these styles. Bilingual training materials, shared dashboards, and co-located coordination meetings fostered mutual understanding.

The partnership with TBEA China demonstrated that when data transparency replaces hierarchical communication, cross cultural friction diminishes and joint accountability strengthens.

A2.7 Lessons Learned

1. **Leadership Commitment Drives Transformation.** Executive participation must remain visible and continuous to sustain momentum.
2. **Governance Ensures Credibility.** Digital audit trails build trust with clients and donors.
3. **Training Builds Capability.** Regular, hands-on workshops convert skepticism into confidence.
4. **Supply Chain Visibility Enhances Predictability.** Integrated dashboards transform delays into manageable variances.
5. **Cross Cultural Trust Multiplies Efficiency.** Shared digital platforms equalize understanding and expectations across international teams.

A2.8 Strategic Implications

CCCE's experience validates the Systems Thinking model developed in this thesis. Its reinforcing feedback loops (R1-R3) operate simultaneously:

- **R1 (Leadership-Training-Efficiency):** Leadership sponsorship amplifies digital competence and efficiency.
- **R2 (Collaboration-Resilience):** Cross Cultural Collaboration fosters trust and adaptive learning.
- **R3 (Supply Chain Visibility):** Real-time logistics integration converts uncertainty into reliability.
- **B1 (Change-Communication-Adaptation):** Continuous communication stabilizes transformation momentum.

Together, these mechanisms demonstrate that digital transformation is not linear but cyclical constantly learning, adapting and reinforcing itself.

A2.9 Conclusion

The CCC Engineering Ltd. case confirms that **leadership commitment, transparent governance, continuous training and supply chain integration** are the essential pillars of strategic digital transformation in the EPC sector. CCCE's Bangladesh operations illustrate how global systems and local realities can be harmonized through technology enabled collaboration.

Ultimately, CCCE's journey exemplifies how a developing economy EPC firm can achieve global standards by institutionalizing leadership, governance and digital supply chain visibility as interconnected strategic capabilities.

Source: Author's compilation based on interview evidence and publicly available company materials (2025).

Appendix 3

Distributed Survey Questionnaire (Used for Quantitative Data Collection)

(Conducted September-October 2025 via Google Forms)

<https://forms.gle/Fez49FiiPfE2Ga4K6>

Survey Title:

Digital Transformation in Strategic Project Management for Exploring Organizational Enablers in Multinational EPC Firms.

Purpose:

The objective of this survey was to collect insights from engineering, project and procurement professionals in multinational EPC organizations located in Finland, Germany and Bangladesh.

The questions examine (a) the current level of digital tool adoption, (b) the role of organizational enablers such as leadership and communication and (c) perceived effects on project efficiency, transparency and resilience.

Participation was voluntary, anonymous and purely for academic research at Tampere University of Applied Sciences (TAMK).

All items used five-point Likert scales (1 = Strongly Disagree / Very Low Impact, 5 = Strongly Agree / Very High Impact) unless otherwise noted.

Section 1. Demographics & Organizational Context

1. What is your gender?

- Male
- Female
- Prefer not to say

2. What is your age group?

- 18–24
- 25–34

- 35–44

- 45–54

- 55 +

3. Country of employment

- Finland
- Germany
- Bangladesh

4. Current role/position in your organization

- C-Level Executive / Director
- Senior Management (Project Lead / Department Head)
- Project Manager / Engineer
- Procurement or Supply Chain Specialist
- IT / Digital Transformation Professional
- Other (please specify)

5. Industry sector of your organization

- Power Transmission / Energy Infrastructure
- Construction & Civil Works
- Industrial Engineering / Manufacturing
- Technology & Automation Services
- Other (please specify)

Section 2. Digital Transformation Practices

6. To what extent are digital tools integrated into your organization's project execution processes?

- Not at all
- Slightly
- Moderately
- Highly integrated
- Fully integrated

7. Which digital tools are used most frequently in your projects? (Select all that apply.)

- ☒ Artificial Intelligence / Machine Learning
- ☒ Blockchain or Smart Contracts
- ☒ Building Information Modelling (BIM)
- ☒ Internet of Things (IoT) / Digital Twins
- ☒ Automation / Robotic Process Automation (RPA)
- ☒ Cloud Based Collaboration Systems
- ☒ Other (please specify)

8. How would you describe your organization's stage of digital transformation?

- Initial pilot phase
- Partial implementation in selected departments
- Moderate integration across projects
- Advanced integration enterprise-wide
- Fully digital / data driven operations

9. Rate the impact of digital tools on the following areas of project performance:

Performance Dimension	1 Very Low	2 Low	3 Moderate	4 High	5 Very High
Efficiency (time & cost control)					
Transparency & accountability					
Resilience & risk management					

Section 3. Organizational Enablers of Digital Transformation

(Please indicate your level of agreement with each statement.)

A. Leadership Commitment

Our top management actively champions digital projects.

Leaders allocate resources and time to digital initiatives.

Visible leadership support improves project execution outcomes.

B. Cross Cultural Collaboration

Our multinational teams communicate effectively through digital platforms.

Cultural differences are openly discussed and managed during projects.

Cross regional cooperation enhances digital innovation.

C. Change Management

Digital implementation in our organization follows a structured change management plan.

Employees are consulted and prepared before new tools are introduced.

Resistance to digital change is handled proactively.

D. Digital Skills & Training

Our organization provides adequate training on digital platforms.
 Employees feel confident using new technologies in projects.
 Continuous learning programs support digital competence development.

E. Communication & Transparency

Project updates and progress are shared openly across departments.
 Digital tools have improved clarity in roles and responsibilities.
 Information flows between HQ and project sites are timely and accurate.

F. Governance & Policy Support

Our organization has clear digital governance policies and data standards.
 Compliance requirements are well aligned with digital processes.
 Top management ensures accountability for data security and ethical use of technology.

Section 4 – Challenges & Future Readiness

10. Which challenges most affect digital transformation in your organization? (Select up to 3)

- ☒ Budget limitations
- ☒ Employee resistance
- ☒ Skills gaps / training needs
- ☒ Leadership inconsistency
- ☒ Poor communication between HQ and sites
- ☒ Weak governance or policies
- ☒ Technological integration issues
- ☒ Other (please specify)

11. How prepared is your organization to scale digital transformation over the next 3 years?

1 Very Low 2 Low 3 Moderate 4 High 5 Very High

12. How confident are you that digital transformation will improve overall project performance in your sector?

1 Not confident 2 Slightly 3 Somewhat 4 Confident 5 Very confident

Section 5. Open Ended Questions

13. In your experience, what has been the most successful aspect of digital transformation in your organization?

14. What do you see as the greatest barrier to effective implementation of digital tools in EPC projects?

15. What organizational change or leadership practice would most help accelerate digital adoption in the future?

Ethical Note

All responses were anonymous and voluntary. No personally identifiable information was collected. Participants were informed of the research purpose and their right to withdraw at any time. Data were used solely for academic analysis within the Master's thesis framework of TAMK Finland.

Appendix 4

Summary of Quantitative Data Tables and Figures

This appendix provides an overview of all quantitative tables and figures presented in Chapter 6 to help readers quickly locate analytical visuals and understand the overall data flow.

All numerical analyses and graphics are interpreted in the main text; therefore, this appendix functions as a navigation reference summarizing the structure of quantitative analysis and its integration into the Systems Thinking synthesis.

A4.1 Overview of Quantitative Data

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A4.2 Quantitative Data Flow

Figure 8 below visually summarizes the analytical progression from raw survey data to integrated systems synthesis.

The figure shows how descriptive statistics, correlation analysis and sentiment interpretation were sequentially combined to construct the Systems Thinking framework that underpins the overall discussion and conclusions.

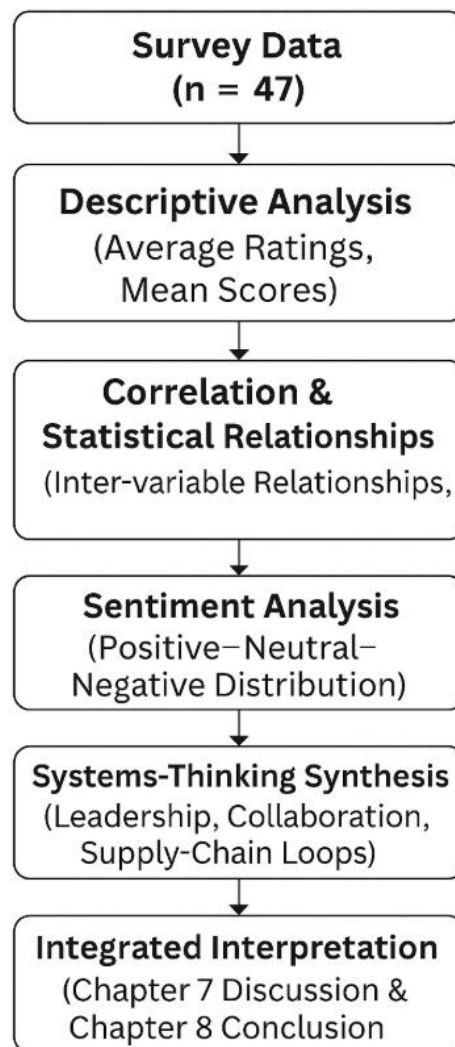


Figure 8. Quantitative Data Flow and Visualization Overview

The figure illustrates the progression from survey data (n = 47) through descriptive analysis, correlation testing, sentiment mapping and systems synthesis, culminating in integrated interpretation across Chapters 7 and 8.

Note

All quantitative visuals were produced using Microsoft Excel and IBM SPSS v.29. Raw survey data (n = 47) are securely archived and available to supervisors upon request for verification or replication.

Appendix 5
Statement Regarding the Use of Artificial Intelligence (AI) in the
Thesis

USE OF AI IN THESIS

I have used AI tools in my thesis:

- ☐ No
☒ Yes

The AI tools used in my thesis and the purpose of their use has been described below:

Names of the tools and versions:

- ChatGPT (GPT-5, OpenAI, 2025 version)
- Grammarly (Version 1.0.87, Premium 7 day trial)

Purpose of the use:

The AI tools were used solely to support academic writing and language refinement. They assisted in improving grammar, sentence clarity, consistency of terminology and the formal academic tone of the text.

No AI tool was used to generate research ideas, analyze data, interpret findings or draw conclusions.

All research design, data collection, data analysis, interpretation and conclusions were entirely performed by the author.

Parts in which AI was used:

AI assistance (ChatGPT and Grammarly) was applied during the editing and proofreading stages of the thesis to:

- Correct grammatical errors and enhance sentence structure;
- Ensure consistency in terminology, especially in the methodology and discussion chapters;
- Refine phrasing for academic clarity and coherence.

The AI tools were not used for generating research content, constructing literature review summaries, analyzing survey or interview data, or creating any visual or numerical results.

I am aware that I am totally responsible for the entire content of the thesis, including the parts generated by AI and i accept the responsibility for any violations of the ethical standards of publications.

Signature:  _____

Name: Somir Roy

Date: November 2025